Orthogonal Rational Basis-Function-Based Discriminant Analysis for Hyperspectral Image Classification

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***Abstract*—This article presents a nonlinear discriminant analysis (NDA) method based on an orthogonal rational basis function (ORF) system for the classification of hyperspectral images (HSIs). The proposed method, abbreviated ORF-NDA, is formulated from the standpoint of function theory. It can make good use of the spectral curve characteristics of HSIs and establish a methodology more suitable for HSI processing in the function space. This is a new paradigm that is different from mainstream hyperspectral data processing methods. First, a functional data fitting model based on an adaptive ORF system is designed for the reconstruction of the spectral curve for each pixel in HSIs. Second, the stochastic adaptive Fourier decomposition (SAFD) method is used to find the optimal parameters for building the adaptive ORF system. Third, based on the reconstructed spectral curve, an NDA method based on an ORF system is built for feature extraction of HSIs. Finally, a spectral–spatial classification framework is designed for HSIs in scenarios with limited sample sizes. Experimental results on five public HSIs demonstrate that the proposed ORF-NDA method outperforms several state-of-the-art deep learning methods in classification accuracy.**

***Index Terms*—Adaptive Fourier decomposition, hyperspectral image (HSI), nonlinear discriminant analysis (NDA), orthogonal rational basis function (ORF).**

I. Introduction

HYPERSPECTRAL imaging acquires rich information for land covers in the ultraviolet, visible, near-infrared, and mid-infrared regions. Unlike RGB images, an HSI records a continuous spectrum for each pixel in the 2-D image. This results in a 3-D (2-D spatial + 1-D spectral) data cube for analysis. The rich information allows for the classification and detection of objects with high precision and detail. This makes hyperspectral imaging particularly useful in a wide range of applications [1], such as precision agriculture, environmental monitoring, mining, and mineralogy. Thus, hyperspectral imaging, with its unique properties, opens up new possibilities in various fields [2].

HSI classification aims at assigning pixels to classes using spectral and spatial features. It can be divided into supervised, semisupervised, or unsupervised based on labeled samples. Classic machine learning methods (e.g., k-nearest neighbor (K-NN), support vector machines (SVM), and random forest (RF) [3]) and recent deep learning models [4], [5] have been applied for classification, but the high dimensionality presents hurdles in terms of computational complexity and the need for extensive training samples. Feature mining (e.g., principal component analysis (PCA), linear discriminant analysis (LDA), convolutional neural network (CNN), and other deep learning methods [6]), spectral–spatial integration, and multimodel fusion are advancing efficient, accurate classification.

Beyond HSI classification's common issues (high dimensionality and limited labeled samples), both classic machine learning and deep learning have unique problems. First, they fail to fully use spectral information [7]. By ignoring spatial information, some advanced methods underperform SVM when few training samples are available. Second, classical approaches rely on feature extraction/selection to reduce dimensionality, which inevitably discards some useful information. Although CNN-based methods [8] can automate feature learning for 2-D or 3-D data, they often treat the spectral dimension in a simplistic manner. Some approaches [9], [10], [11], [12] apply PCA or band selection to obtain feature maps, which emphasize spatial over spectral discriminability. Third, an acute problem is the computational costs. Classic machine learning methods often have significant advantages over deep learning methods. Especially in some large-scale HSIs, many deep learning models cannot be executed due to insufficient memory. This is mainly because the patch-splitting operation involved in these models generates a memory bottleneck and a large amount of computing time [13]. Although

Received 3 December 2025; revised 12 February 2026; accepted 13 May 2026. Date of publication 19 May 2026; date of current version 28 May 2026. This work was supported in part by the Science and Technology Development Fund of Macau Special Administrative Region (SAR) under Grant 0079/2023/ITP2, Grant 0097/2025/ITP2, and Grant 0041/2023/RIA1; in part by Wuyi University Hong Kong-Macau Joint Research and Development Fund under Grant 2022WGALH16; and in part by the National Natural Science Foundation of China under Grant 42471417. *(Corresponding author: Liming Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3694091

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patch-based methods can capture local structural features more easily, the amount of calculation and memory requirements they bring is also significant.

Transformers [14] have achieved great success by using self-attention to capture contextual relations along input sequences, illustrating that the order of data matters in sequences. Spectral data exhibit a similar sequential property, yet many methods treat HSIs as simple reflectance vectors, ignoring band order. In recent years, researchers have used Transformers to model spectral sequences for HSI classification. Hong et al. [15] presented a novel backbone network from a sequential perspective with Transformer (SpFormer). Several works have fused CNN-based spatial features with Transformer-based spectral modeling to capture both local spatial information and long-range spectral dependencies. For instance, Sun et al. [16] proposed a spectral–spatial feature tokenization transformer (SSFTT) method integrating a backbone CNN and transformer structure for HSI classification. Tian et al. [17] designed a max-gap loss-guided Siamese reconstruction network (MAGS) for HSI classification. Shen et al. [18] proposed a simple multilayer-perceptron (MLP)-based model named SimHSIC for HSI classification. In summary, Transformers are widely recognized for their efficient sequential data processing via self-attention mechanisms. However, their utilization of the continuous characteristics of HSI data always remains at the vector level. Hyperspectral imaging captures spectral data at a high resolution, resembling continuous curves. As the resolution improves, these curves become more complete and continuous, correlating with wavelength. Therefore, HSIs fundamentally represent functional data (high-resolution spectral curves) with spatial structure. In this article, we analyze hyperspectral data from the standpoint of function theory, aiming to explore efficient methods for HSI processing. This approach allows us to capture the continuous nature of the spectral data, which is essential for understanding the inherent features of HSIs.

Functional data analysis is a developing statistical methodology. It deals with the analysis and theory of continuous data in the form of functions [19]. The relevant theory and methods have been widely used in some fields [20], [21]. Functional principal component analysis (FPCA) is the most representative method. The data characteristics of HSIs are very suitable for processing using functional data analysis methods. Many attempts have been made in this regard, and some achievements have been accomplished. For example, Li et al. [22] extracted the functional spectral features of HSI data by FPCA for classification. Xue et al. [23] presented a set of improved FPCA methods for HSI classification by introducing supervised and semisupervised mechanisms. Seifi Majdar and Ghassemian [24] proposed a spectral–spatial classification method using FPCA and a watershed segmentation algorithm. Lv et al. [25] also proposed a spatial FPCA method for HSI classification. Furthermore, Zullo et al. [26] proposed the functional extension of SVM and RF based on FPCA for HSI classification. These methods achieved excellent accuracies. However, all the works are mainly based on the very mature FPCA based on the B-spline function. Because of that, there are three significant problems. First, unsupervised FPCA cannot use label information to learn discriminative features. Second, B-spline functions in FPCA can obtain better approximation effects. However, its convergence speed is too slow and usually requires more basis functions than the dimension of the data. Third, the B-spline basis is nonorthogonal, leading to representation coefficients that lack uniqueness and completeness. Consequently, using these coefficients directly as features typically leads to markedly degraded classification performance. Thus, the methodology of functional data analysis for HSI processing needs to be further developed. New methods need to be proposed to enrich the existing theory and algorithms in HSI classification.

To mitigate the challenges of high dimensionality and insufficient exploitation of spectral sequence data, we propose here orthogonal rational basis function (ORF)-NDA, a new nonlinear discriminant analysis (NDA) method from a functional perspective. ORF-NDA leverages an ORF system for NDA to model spectral data as continuous functions rather than discrete vectors, as shown in Fig. 1. This novel method transitions from the handling of vector-based data to a function-based paradigm encapsulating the continuous spectral characteristics inherent in HSI data. The ORF-NDA can effectively yield compact, identifiable coefficients and extract discriminative features that capture both local and long-distance dependencies across bands. First, a new functional data fitting model based on an adaptive ORF system is designed to reconstruct the continuous spectral curve for each pixel in the function space. Second, a stochastic adaptive Fourier decomposition (SAFD) method is introduced to adaptively select an appropriate ORF system based on the characteristics of HSI data. Third, based on the reconstructed spectral curve, an NDA method is proposed for functional feature extraction. Finally, a spectral–spatial classification framework based on simple mean filtering is designed for the classification of land covers. This proposed method combines the ideas of functional data analysis and machine learning and is suitable for high-dimensional continuous data analysis and processing. The main contributions of this article consist of the following.

1) To address high dimensionality and the insufficient utilization of spectral data, a highly efficient functional data fitting model is constructed to recover the underlying spectral curve for each pixel. This treats the spectral function as the fundamental unit, aligning better with the data's continuous nature and departing from mainstream vector-based HSI processing pipelines.
2) Fixed-basis function systems lack the flexibility to reflect HSI-specific, dataset-dependent spectral patterns. Our SAFD-driven method dynamically constructs the optimal ORF systems tailored to each HSI dataset. This adaptability ensures high-precision curve reconstruction with a few terms. The coefficient energy exhibits fast, near-monotonic decay, confirming the compactness and efficiency of the functional representation.
3) By leveraging the continuity and orthogonality of the ORF-based spectral functions, a new NDA method, NDA, is proposed for functional feature extraction of

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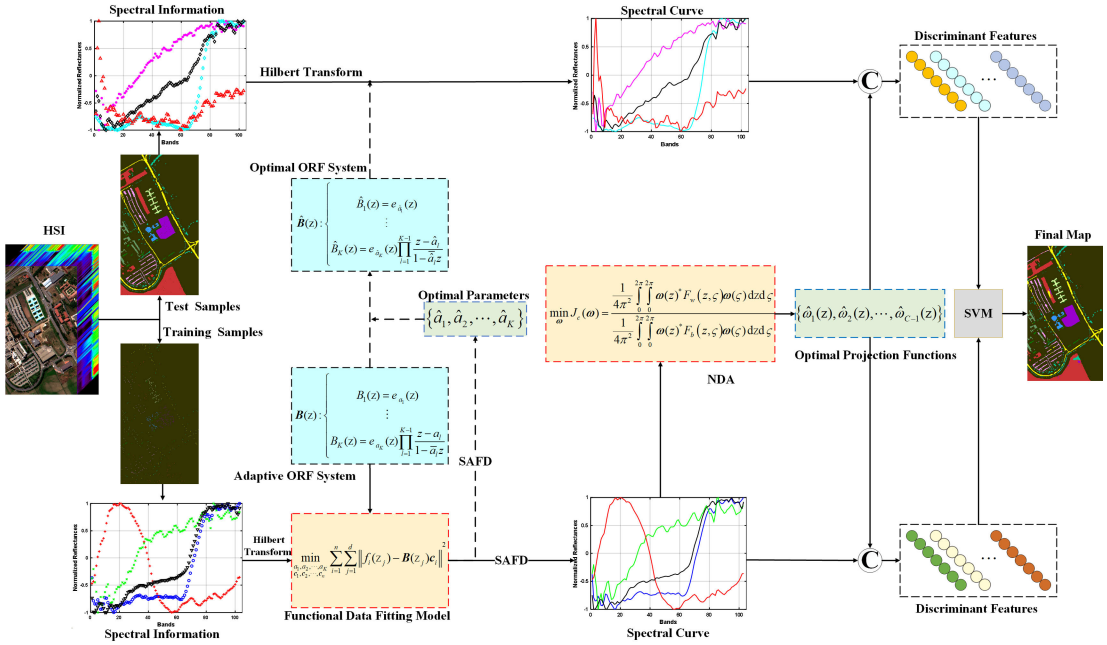


Fig. 1. Framework of the proposed ORF-NDA, including the ORF system, the functional data fitting model, SAFDmethod, and the NDA in the function space. The ORF system can adaptively construct the optimal basis function system based on the training samples, as shown by the dotted line. The functional data fitting model is based on an adaptive basis function system for all training samples. Since the basis functions are not determined, the explanation of this problem requires the use of SAFD to gradually determine the parameters $\{a_k\}_{k=1}^{K}$ of the basis functions. The NDA is specially designed to mine the discriminative features of spectral curves (functions) in the function space. It establishes discriminant analysis criteria in the function space to find the best projection function and achieve the aggregation and dispersion of spectral curves. The symbol © represents the Cauchy integral.

infinite-dimensional spectral curves. The reproducing property, normalization, and orthogonality of the basis functions are utilized to significantly simplify the difficulty of solving the discriminant model in the function space.

4) To utilize spatial information while maintaining spectral continuity, a spectral–spatial classification framework based on ORF-NDA and common mean filtering is presented. We first apply common mean filtering to form initial spectral–spatial features, then use ORF-NDA to extract compact functional features for classification. This proposed framework shows robust performance even with limited training samples.

The article is divided into four sections. Section II mainly describes the functional data fitting model, construction of the ORF system, the SAFD method, NDA for spectral curves, and a spectral–spatial classification framework based on mean filtering. Section III presents experimental outcomes on five HSI datasets, along with a comparative analysis between the proposed method and various state-of-the-art methods. Finally, Section IV offers concluding remarks.

II. Mathematical Methodology

Each pixel within an HSI contains hundreds of reflectance values of specific bands. As spectral resolution increases, each pixel can generate a comprehensive and continuous spectral curve. Consequently, HSI data possesses functional characteristics. Thus, we consider converting HSI data into spectral curves and processing them from a functional perspective.

A. Functional Data Fitting Model

The spectral vector of each pixel can be viewed as individual sampling points along a functional curve. The basis function representation method can effectively reconstruct the potential spectral curve. To achieve this, we first treat the spectral vector as functional data, providing a solid foundation for further analysis and processing.

Let $\boldsymbol{X} \in \mathbb{R}^{n\times d}$ be an HSI with n pixels $\{\boldsymbol{x}_i, y_i\}_{i=1}^{n}$, where $\boldsymbol{x}_i \in \mathbb{R}^d$ is the spectral data for the ith pixel, and y_i is the potential label in the class label set $\boldsymbol{\Omega} = \{1, \ldots, C\}$. From a functional perspective, the elements of $\boldsymbol{x}_i = [\boldsymbol{x}_{i1}, \boldsymbol{x}_{i2}, \ldots, \boldsymbol{x}_{id}]^T$ are sampled sequentially from the corresponding spectral curve $x_i(t)$, where t denotes the wavelength. In the actual data acquisition process, HSIs inevitably contain observation errors. Based on this principle, the raw spectral data can be expressed as follows:

$$\boldsymbol{x}_{ij} = x_i\left(t_j\right) + r_i\left(t_j\right), \quad j = 1, 2, \ldots, d. \tag{1}$$

To accurately reconstruct the original spectral curve $x_i(t)$, we hope to minimize the energy of the error function (remainder) $r_i(t)$ at given wavelengths by the l^2-norm. Thus, we can establish the following functional data fitting model:

$$\min_{x_i} \left\|\varepsilon_i(t)\right\|_2^2 = \min_{x_i} \sum_{j=1}^{d} \left[\boldsymbol{x}_{ij} - x_i\left(t_j\right)\right]^2. \tag{2}$$

The above model aims to find the optimal spectral function $x_i(t)$. However, since the specific form of the function is not known, this model is difficult to handle. Moreover, the spectral function $x_i(t)$ may be a very complex expression or

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cannot be expressed with explicit formulas. Therefore, we can employ the basis function representation method to gradually approximate the spectral function $x_i(t)$. According to the Stone–Weierstrass theorem, any continuous function defined on a closed and bounded interval can be approximated with arbitrary precision using a polynomial or trigonometric basis function system [27]. Thus, the basis function representation method provides us with an effective idea for reconstructing spectral functions. The key idea is to express $x_i(t)$ as a linear combination of the basis functions, i.e.,

$$x_i(t) = \sum_{k=1}^{K} c_{ik}\phi_k(t) = \boldsymbol{c}_i^T \Phi(t) = \Phi(t)^T \boldsymbol{c}_i \tag{3}$$

where $\Phi(t) = [\phi_1(t), \phi_2(t), \ldots, \phi_K(t)]^T$ is the basis function system with K elements, and $\boldsymbol{c}_i = [c_{i1}, c_{i2}, \ldots, c_{iK}]^T$ is the corresponding functional representations. The parameter K can theoretically approach an infinite positive integer, but in practical applications, we hope to use the smallest possible positive integer K to achieve the ideal approximation effect. The common basis function systems mainly include the polynomial basis system, the Fourier basis system, and the B-spline basis system. According to the property of basis functions, the corresponding basis function system has its range of applications. For example, the Fourier basis system is suitable for periodic data, while the B-spline basis system is suitable for open-ended data. Logically speaking, the B-spline basis function system is a very good choice for reconstructing spectral functions. However, it faces several key problems. First, the construction of the B-spline basis function is fixed and only related to nodes, resulting in limited flexibility. Second, the number of basis functions is large, and the convergence speed is slow. At the same time, in order to achieve better results, B-spline basis function systems usually require more basis functions, and their number is closely related to the data dimension. Finally, the nonorthogonality of B-spline basis functions means that there may be a certain correlation between different B-spline basis functions, rather than being completely independent. This can lead to increased computational complexity and numerical stability issues.

ORF systems have significant advantages in signal processing and image decomposition [28], [29], [30]. In this article, an ORF system based on Takenaka–Malmquist (TM) basis functions is introduced for reconstructing spectral functions. The TM system can approximate any nonlinear function with a fast learning convergence speed and has good generalization ability. Moreover, the TM system can find an adaptive approximation in the energy sense by selecting the optimal parameters for constructing rational orthogonal basis functions according to the characteristics of the spectral curves in HSIs. The TM system is built based on the normalized Szegö kernel functions [31] by the Blaschke products in the complex Hardy space $H^2(\mathbb{D})$, where $\mathbb{D} \triangleq \{re^{\mathbf{i}t} \in \mathbb{C} : 0 \le r < 1\}$ denotes the open unit disk centered on the origin in the complex plane $\mathbb{C}$ and the symbol $\mathbf{i}$ is the imaginary unit.

First, the normalized Szegö kernel functions, with the variable $z = e^{\mathbf{i}t}$ defined on the unit circle ∂D^+ following the anticlockwise direction around D, can be defined as follows:

$$e_{a_k}(z) = \frac{\sqrt{1-|a_k|^2}}{1-\bar{a}_k z}, \quad a_l \in \mathbb{D} \quad \& \quad k = 1, 2, \ldots, K. \tag{4}$$

Here $\bar{a}_k$ is the complex conjugate of a_k. Note that the normalized Szegö kernel functions have the following property under the Cauchy integral:

$$\left\langle e_{a_k}, e_{a_k} \right\rangle = \frac{1}{2\pi}\int_0^{2\pi} \frac{1-|a_k|^2}{\left(1-\bar{a}_k e^{\mathbf{i}t}\right)\left(1-a_k e^{-\mathbf{i}t}\right)} dt = 1. \tag{5}$$

Then, the TM system $\{B_k(z)\}_{k=1}^K$ is built by the Blaschke products

$$B_k(z) = \begin{cases} e_{a_1}(z), & k = 1 \\ e_{a_k}(z)\prod_{l=1}^{k-1} \frac{z-a_l}{1-\bar{a}_l z}, & k = 2, 3, \ldots, K. \end{cases} \tag{6}$$

By using the Cauchy integral formula in [28], the following three important properties can be proved.

1) *Reproducing Property*

$$\left\langle f, e_{a_k} \right\rangle = \sqrt{1-|a_k|^2} f(a_k) \quad \forall f \in H^2\left(\partial D^+\right). \tag{7}$$

2) *Normalization*

$$\langle B_k, B_k \rangle = 1, \quad k = 1, 2, 3, \ldots \tag{8}$$

3) *Orthogonality*

$$\langle B_k, B_l \rangle = 0 \quad \forall k \ne l. \tag{9}$$

Thus, the TM system is an orthonormal system, which depends on the choices of the complex parameters $\{a_k\}_{k=1}^K \subset D$. In particular, when $a_1 = a_2 = \cdots = a_K = 0$, this TM system becomes the Fourier basis function system $\{z^k\}_{k=0}^K (z = e^{\mathbf{i}t})$. The adaptive Fourier decomposition (AFD) method [28] aims at searching the optimal parameters $\{a_k\}_{k=1}^K$ to construct the optimal TM system for different signals.

Considering that an HSI is a collection of real signals, we first convert the real signals into the complex domain through the Hilbert transform $\mathcal{H}$ [32] and then use the related tools of the complex variable function for processing. This methodology is also different from mainstream HSI processing approaches. To establish the analytical function space $H^2(\mathbb{D})$ on the unit disk, the wavelength t is transformed to $[0, 2\pi]$. The analytic signal $f_i(z)$ corresponding to any real signal $x_i(t)$ in an HSI can be obtained by the Hilbert transform $(z = e^{\mathbf{i}t})$

$$f_i(z) = \frac{1}{2}\left[x_i + \mathbf{i}\mathcal{H}(x_i) + c_{i0}\right], c_{i0} = \frac{1}{2\pi}\int_0^{2\pi} x(t)\, dt. \tag{10}$$

The spectral analytic signal $f_i(z)$ for $\boldsymbol{x}_i$ under the TM system in (6) can be represented as follows:

$$f_i(z) = \sum_{k=1}^{K} c_{ik} B_k(z) + r_i(z) = \boldsymbol{B}(z)\boldsymbol{c}_i + r_i(z) \tag{11}$$

where

$$\boldsymbol{B}(z) = [B_1(z), B_2(z), \ldots, B_K(z)] \tag{12}$$

and

$$\boldsymbol{c}_i = [c_{i1}, c_{i2}, \ldots, c_{iK}]^T. \tag{13}$$

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For any analytic signal $f_i(z)$, the functional data fitting model in (2) based on the TM system can be rewritten as follows:

$$\min_{a_1,a_2,\ldots,a_K} \|r_i(z)\|_2^2 = \min_{a_1,a_2,\ldots,a_K} \sum_{j=1}^{d} \left[f_i(z_j) - \boldsymbol{B}(z_j)\boldsymbol{c}_i\right]^2 = \min_{a_1,a_2,\ldots,a_K} \|f_i - \mathbb{B}\boldsymbol{c}_i\|_2^2. \quad (14)$$

Here $z_j = e^{\mathbf{i}t_j}$ and $\mathbb{B}$ is the basis function matrix defined as follows:

$$\mathbb{B} = \begin{pmatrix} \boldsymbol{B}(z_1) \\ \boldsymbol{B}(z_2) \\ \vdots \\ \boldsymbol{B}(z_K) \end{pmatrix} = \begin{pmatrix} B_1(z_1) & B_2(z_1) & \cdots & B_K(z_1) \\ B_1(z_2) & B_2(z_2) & \cdots & B_K(z_2) \\ \vdots & \vdots & \ddots & \vdots \\ B_1(z_d) & B_2(z_d) & \cdots & B_K(z_d) \end{pmatrix}. \quad (15)$$

If the basis function matrix $\mathbb{B}$ is known, the above problem can be easily solved using the least squares method. Due to the orthogonality property in (9) of the TM system, the representation coefficient c_{ik} can be defined by the Cauchy integral as in (5)

$$c_{ik} = \langle f_i, B_k \rangle = \frac{1}{2\pi} \int_0^{2\pi} f_i\left(e^{\mathbf{i}t}\right) e^{-\mathbf{i}kt} dt. \quad (16)$$

However, according to the defined formula in (6), we know that the TM system is flexible and changeable, and it is related to the parameters $\{a_k\}_{k=1}^K$. Furthermore, according to the analytic signals $\{f_i\}_{i=1}^n$ derived by an HSI, a functional data fitting model based on the TM system for n spectral signals is built in the complex domain

$$\min_{a_1,a_2,\ldots,a_K} \sum_{i=1}^{n} \|r_i(z)\|_2^2 = \min_{a_1,a_2,\ldots,a_K} \sum_{i=1}^{n} \|f_i - \mathbb{B}\boldsymbol{c}_i\|_2^2. \quad (17)$$

To solve the model in (17), we first need to find the optimal parameters $\{\hat{a}_k\}_{k=1}^K$ and construct the optimal TM system for all spectral signals in an HSI. Then the representation coefficient $\boldsymbol{c}_i$ for each sample is easily obtained based on (16), and the spectral curves can be precisely represented by the line combination of the TM system.

B. Stochastic Adaptive Fourier Decomposition

SAFD [30] is a statistical version of AFD, which aims to select a common set of atoms from the Szegö kernel dictionary to represent a group of signals simultaneously. This is highly consistent with the function data fitting model in (15) that we want to solve. Here, we need to seek a group of basis functions for multiple spectral signals according to the spectral characteristics in different HSIs. Therefore, we will employ the idea of SAFD to construct the optimal TM system.

Based on the model in (17), a stepwise optimization strategy is adopted to sequentially seek the optimal parameters $\hat{a}_1, \hat{a}_2, \ldots, \hat{a}_K$. Moreover, at each step, we use the energy minimization principle of the remainders to select the optimal parameters. The specific steps are as follows.

First, to seek the optimal value $\hat{a}_1$ of the unknown parameter a_1, we can rewrite (11) as follows:

$$f_i(z) = c_{i1}B_1(z) + \sum_{k=2}^{K} c_{ik}B_k(z) \triangleq c_{i1}e_{a_1}(z) + r_{i1}(z) \quad (18)$$

where $r_{i1}(z) = \sum_{k=2}^K c_{ik}B_k(z)$ is the first remainder for the ith spectral signal. According to the energy minimization principle and (18), an alternative and simplified optimization model for (17) at the first step can be built as follows:

$$\hat{a}_1 = \arg\min_{a_1} \sum_{i=1}^{n} \|r_{i1}(z)\|_2^2 = \arg\max_{a_1} \left(1 - |a_1|^2\right) \sum_{i=1}^{n} |f_i(a_1)|^2. \quad (19)$$

Note that in (19), the third step can be derived from the second step by the normalization, (8), and orthogonality, (9), of the TM system $\{B_k(z)\}_{k=1}^K$. The fourth step is an equivalent optimization problem of the third step, with the total energy finite and the equality $\|f_i\|_2^2 = |c_{i1}|^2 + \sum_{k=2}^K |c_{ik}|^2$. The fifth step is derived based on (16). The final step can be derived from the reproducing property in (7). Because $a_1 \in \mathbb{D}$ and the wavelength t is transformed to $[0, 2\pi]$, the optimal parameter $\hat{a}_1$ can be quickly searched based on the final equation in (1). The optimal representation coefficient $\hat{c}_{i1}$ and basis function $\hat{B}_1(z)$ corresponding to c_{i1} and $B_1(z)$ can be obtained by

$$\hat{B}_1 = e_{\hat{a}_1}(z) = \frac{\sqrt{1 - |\hat{a}_1|^2}}{1 - \bar{\hat{a}}_1 z} \quad (20)$$

and

$$\hat{c}_{i1} = \sqrt{1 - |\hat{a}_1|^2} f_i(\hat{a}_1). \quad (21)$$

The spectral curves can be expanded to the following form:

$$f_i(z) = \hat{c}_{i1}\hat{B}_1(z) + r_{i1}(z), \quad i = 1, 2, \ldots, n. \quad (22)$$

Second, we can seek the optimal parameter $\hat{a}_2$ based on the resulting optimal parameter $\hat{a}_1$. The first remainder for the ith spectral signal $f_i(z) \triangleq f_i^{+1}(z)$ can be accurately expressed after obtaining the optimal parameter $\hat{a}_1$

$$r_{i1}(z) = f_i(z) - \hat{c}_{i1}\hat{B}_1(z) = f_i^{+1}(z) - \frac{1 - |\hat{a}_1|^2}{1 - \bar{\hat{a}}_1 z} f_i(\hat{a}_1). \quad (23)$$

We can verify that $z = \hat{a}_1$ is a zero point of $r_{i1}(z)$ in (23) due to $r_{i1}(\hat{a}_1) = 0$. Next, we use an Möbius transformation $\tau_a(z)$ with the parameter $a = \hat{a}_1$ to eliminate the zero point $z = \hat{a}_1$ of the first remainder $r_{i1}(z)$ and obtain a new form, denoted by $f_i^{+2}(z)$

$$f_i^{+2}(z) = \frac{r_{i1}(z)}{\tau_{\hat{a}_1}(z)} = \frac{f_i(z) - \hat{c}_{i1}\hat{B}_1(z)}{\frac{z - \hat{a}_1}{1 - \bar{\hat{a}}_1 z}}. \quad (24)$$

By implementing the identity transformation, (25) can be transformed into

$$f_i(z) = \hat{c}_{i1}\hat{B}_1(z) + f_i^{+2}(z)\frac{z - \hat{a}_1}{1 - \bar{\hat{a}}_1 z}. \quad (25)$$

Similar to $f_i(z)$ in (18), $f_i^{+2}(z)$ is also expanded to the following form (26) by a newly normalized Szegö kernel function $e_{a_2}(z)$ defined in (4) with the unknown parameter a_2

$$f_i^{+2}(z) = c_{i2}e_{a_2}(z) + r_{i2}(z) \quad (26)$$

where $r_{i1}(z)$ is the second remainder for the ith spectral signal. According to the energy minimization principle, the

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optimization model for n spectral signals at the second step can be built like in (19) as follows:

$$\hat{a}_2 = \arg\min_{a_2} \sum_{i=1}^{n} \|r_{i2}(z)\|_2^2 = \arg\max_{a_2} \left(1-|a_2|^2\right) \sum_{i=1}^{n} \left|f_i^{+2}(a_2)\right|^2. \tag{27}$$

After obtaining the optimal parameter $\hat{a}_2$, (25) can be further expanded to the following form by substituting (26) and $\hat{a}_2$

$$f_i(z) = \hat{c}_{i1}\hat{B}_1(z) + \hat{c}_{i2}e_{\hat{a}_2}(z)\frac{z-\hat{a}_1}{1-\hat{a}_1 z} + r_{i2}(z)\frac{z-\hat{a}_1}{1-\hat{a}_1 z} \tag{28}$$

where

$$\hat{c}_{i2} = \sqrt{1-|\hat{a}_2|^2} f_i^{+2}(\hat{a}_2). \tag{29}$$

By setting $\hat{B}_2(z) = e_{\hat{a}_2}(z)(z-\hat{a}_1)/(1-\hat{a}_1 z)$, which is consistent with the definition in (6), (28) can be denoted as follows:

$$f_i(z) = \hat{c}_{i1}\hat{B}_1(z) + \hat{c}_{i2}\hat{B}_2(z) + r_{i2}(z)\frac{z-\hat{a}_1}{1-\hat{a}_1 z}. \tag{30}$$

Repeat: We can also verify that $z = \hat{a}_2$ is a zero point of $r_{i2}(z)$ in (26) due to $r_{i2}(\hat{a}_2) = f_i^{+2}(\hat{a}_2) - \hat{c}_{i2}\hat{B}_2(\hat{a}_2) = 0$. Thus, we can redefine $f_i^{+3}(z)$ similar to $f_i^{+2}(z)$ in (24), and then establish the optimization model like (27) for n spectral signals at the third step to find the optimal parameter $\hat{a}_3$. Equation (30) can be decomposed into

$$f_i(z) = \sum_{k=1}^{3} \hat{c}_{ik}\hat{B}_k(z) + r_{i3}(z)\frac{z-\hat{a}_2}{1-\hat{a}_2 z}\frac{z-\hat{a}_1}{1-\hat{a}_1 z} \tag{31}$$

where

$$\hat{c}_{i3} = \sqrt{1-|\hat{a}_3|^2} f_i^{+3}(\hat{a}_3) \tag{32}$$

and

$$\hat{B}_3(z) = e_{\hat{a}_3}(z)\frac{z-\hat{a}_2}{1-\hat{a}_2 z}\frac{z-\hat{a}_1}{1-\hat{a}_1 z}. \tag{33}$$

By repeating this process, we can find the optimal parameters $\hat{a}_3$, $\hat{a}_4$, $\hat{a}_5$, …, $\hat{a}_K$ in turn.

Finally, based on the resulting optimal parameters $\{\hat{a}_k\}_{k=1}^{K}$, the optimal TM system $\{\hat{B}_k(z)\}_{k=1}^{K}$ can be built for n given spectral signals in an HSI by substituting $\{\hat{a}_k\}_{k=1}^{K}$ into (6). The final expansion equation for (31) can be denoted as follows:

$$f_i(z) = \sum_{k=1}^{K} \hat{c}_{ik}\hat{B}_k(z) + r_i(z) \tag{34}$$

with the final remainder

$$r_i(z) = r_{iK}(z)\prod_{l=1}^{K-1}\frac{z-\hat{a}_l}{1-\bar{\hat{a}}_l z}, \quad K>1. \tag{35}$$

According to the energy minimization principle and the total energy finite $\|f_i\|_2^2 < +\infty$, the remainders have less and less energy with K increasing. Thus, the final remainder satisfies $\|r_i(z)\|_2^2 \to 0$ when $K \to \infty$. Since the aforementioned process gradually optimizes each step to minimize the residual energy, a small value of K can lead to a fast approximation with a sufficiently minor error generated by the final remainder $r_i(z)$ in (34). Thus, by selecting an appropriate value of K, we can establish an optimal basis function system, and enable the reconstruction of the spectral curves with high precision in the complex space, i.e.,

$$\hat{f}_i(z) \approx \sum_{k=1}^{K} \hat{c}_{ik}\hat{B}_k(z) = \hat{\boldsymbol{B}}(z)\hat{\boldsymbol{c}}_i, \quad i=1,2,\ldots,n \tag{36}$$

and the corresponding real signal can also be expressed based on (10)

$$\hat{x}_i(t) = 2Re\left(\hat{f}_i(z)\right) - \hat{c}_{i0}, \quad i=1,2,\ldots,n. \tag{37}$$

Here $\hat{\boldsymbol{B}}(z)$ is the optimal form of (12), and $\hat{\boldsymbol{c}}$ is the corresponding form of (13) under the optimal TM system. Again, let us define $\hat{\mathbb{B}}$ as the optimal form of (15). The algorithmic procedure of the functional data fitting model using SAFD is given in Algorithm 1. Next, we design a new feature extraction method in a complex function space to extract the effective features of the spectral curves.

Algorithm 1 Functional Data Fitting Model Using SAFD

Input: Considering n pixels $\{\boldsymbol{x}_i, y_i\}_{i=1}^{n}$ in given HSI $\boldsymbol{X}$, the randomly initialized parameter set $\{a_k\}_{k=1}^{\mathcal{K}} \subset D$ and the number of basis functions K ($\mathcal{K} \gg K$).

Output: The optimal parameters $\{\hat{a}_k\}_{k=1}^{K}$, the optimal TM system $\{\hat{B}_k(z)\}_{k=1}^{K}$, the representation coefficient vectors $\{\hat{c}_k\}_{k=1}^{K}$, the reconstructed spectral curve $f_i(z)$, and the reconstructed error $\|r_i\|$.

```
Obtain the analytic form f_i of each spectral signal x_i by the Hilbert transform (10).
Initialize f_i^{1+} = f_i, z = e^{it}, t ∈ {2jπ/d}_{j=0}^{d}, and f̂_i(z) = 0.
for k = 1 : K do
    Find â_k = arg max_{a∈{a_k}_{k=1}^{K}} (1 − |a|^2) Σ_{i=1}^{n} |f_i^{+k}(a)|^2.
    Construct the basis function B̂_k(z) by (6) after substituting the parameters {â_1, â_2, ··· , â_k}.
    for i = 1 : n do
        Calculate the representation coefficient ĉ_ik = ⟨f_i, B̂_k⟩.
        Perform f_i^{+(k+1)}(z) = (f_i^{+k}(z) − ĉ_ik B̂_k(z))/ ((z−â_k)/(1−ā_k z)) in an Möbius transformation.
        Calculate f̂_i(z) = f̂_i(z) + ĉ_ik B̂_k(z).
        if k == K then
            Calculate the error ||r_i(z)||_2^2 based on (34).
            Set the coefficient vector ĉ_i = [ĉ_i1, ĉ_i2, ··· , ĉ_iK].
        end if
    end for
end for
Return {â_k}_{k=1}^{K}, {B̂_k(z)}_{k=1}^{K}, {ĉ_i}_{i=1}^{n}, {f̂_i(z)}_{i=1}^{n}, and {||r_i(z)||_2^2}_{i=1}^{n}.
```

C. NDA in Complex Function Space

Discriminant analysis is a statistical method used to solve feature extraction and classification problems. Its principle is to select the projection direction to maximize the similarity between samples of the same class and the difference between samples of different classes. This article establishes an NDA method for functions in the complex space with the help of the idea of discriminant analysis. Based on the resulting spectral

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curves (36), NDA aims to find nonlinear decision boundaries that separate different classes in the complex function space. This boundary is typically denoted by a set of complex-valued functions.

Assuming that there are C classes in this HSI $\boldsymbol{X}$, each class contains n_j labeled samples, denoted as $\{\boldsymbol{x}_i^j, y_i^j\}_{i=1}^{n_j}$, where $j = 1, 2, \ldots, C$ and $m = \sum_{j=1}^{C} n_j$. We can obtain the spectral curves $\{\hat{f}_i^j, y_i^j\}_{i=1}^{n_j}$ of the labeled samples $\{\boldsymbol{x}_i^j, y_i^j\}_{i=1}^{n_j}$ per class according to (36) and $\hat{\boldsymbol{c}}_i^j$ is the coefficient vector of $\hat{f}_i^j$. The mean function (center curve) of the spectral curves for all labeled samples in $\boldsymbol{X}$ can be denoted as follows:

$$u(z) = \frac{1}{m}\sum_{j=1}^{C}\sum_{i=1}^{n_j}\hat{f}_i^j(z) = \frac{1}{m}\sum_{j=1}^{C}\sum_{i=1}^{n_j}\hat{\boldsymbol{B}}(z)\,\hat{\boldsymbol{c}}_i^j = \hat{\boldsymbol{B}}(z)\,\boldsymbol{u}_c. \quad (38)$$

Here, $\boldsymbol{u}_c = 1/m\sum_{j=1}^{C}\sum_{i=1}^{n_j}\hat{\boldsymbol{c}}_i^j$ is the mean vector of representation coefficient vectors for all labeled samples in the complex function space. Based on (38), we can build the mean function $u_j(z)$ of the spectral curves for each class ($j = 1, 2, \ldots, C$) as follows:

$$u_j(z) = \hat{\boldsymbol{B}}(z)\,\boldsymbol{u}_c^j, \quad \boldsymbol{u}_c^j = \frac{1}{n_j}\sum_{i=1}^{n_j}\hat{\boldsymbol{c}}_i^j. \quad (39)$$

Furthermore, the total scatter function $F_t(z,\zeta)$ and the between-class scatter function $F_b(z,\zeta)$ can be built in the complex function space by the definition of the covariance function and the conjugate (Hermitian) transpose operation *

$$\begin{aligned} F_t(z,\zeta) &= \sum_{j=1}^{C}\sum_{i=1}^{n_j}\left(\hat{f}_i^j(z) - \mu(z)\right)\left(\hat{f}_i^j(\zeta) - \mu(\zeta)\right)^* \\ &= \hat{\boldsymbol{B}}(z)\,\boldsymbol{S}_t^c\,\hat{\boldsymbol{B}}(\zeta)^* \\ F_b(z,\zeta) &= \sum_{j=1}^{C}\left(u_j(z) - \mu(z)\right)\left(u_j(\zeta) - \mu(\zeta)\right)^* \\ &= \hat{\boldsymbol{B}}(z)\,\boldsymbol{S}_b^c\,\hat{\boldsymbol{B}}(\zeta)^* \end{aligned} \quad (40)$$

where

$$\begin{aligned} \boldsymbol{S}_t^c &= \sum_{j=1}^{C}\sum_{i=1}^{n_j}(\hat{\boldsymbol{c}}_i^j - \boldsymbol{u}_c)(\hat{\boldsymbol{c}}_i^j - \boldsymbol{u}_c)^* \\ \boldsymbol{S}_b^c &= \sum_{j=1}^{C}(\boldsymbol{u}_c^j - \boldsymbol{u}_c)(\boldsymbol{u}_c^j - \boldsymbol{u}_c)^*. \end{aligned} \quad (41)$$

According to the above derivation, we find that the total scatter function $F_t(z,\zeta)$ is related to the total scatter matrix s_t^c of the representation coefficient vectors in the complex space. Like the total scatter matrix $\boldsymbol{S}_t^c$, the total scatter function $F_t(z,\zeta)$ measures the scattering and clustering between all spectral curves $\{f_i^j, y_i^j\}_{i=1}^{n_j}$ and the center curve $u(z)$. Similarly, as the between-class scatter s_b^c, the between-class scatter function $F_b(z,\zeta)$ measures the scattering and clustering among the center curves $\{u_j(z)\}_{j=1}^{C}$ for different classes. Combining the total scatter function $F_t(z,\zeta)$ and the between-class scatter function $F_b(z,\zeta)$, we can obtain the within-class scatter function $F_w(z,\zeta)$ as follows:

$$F_w(z,\zeta) = F_t(z,\zeta) - F_b(z,\zeta). \quad (42)$$

According to the idea of common discriminant analysis and classification, we hope to achieve the following goals, i.e., the intraclass divergence is as small as possible (clustering). In contrast, the interclass divergence is as large as possible (scattering). In order to achieve this goal, we need to find the optimal projection function to map all spectral curves into some subspace to achieve dispersion and aggregation of all samples according to their classes. Setting any projection function $\omega(z)$ in a complex function space spanned by the optimal TM system $\{\hat{B}_k(z)\}_{k=1}^{K}$, we can denote $\omega(z)$ as follows:

$$\omega(z) = \sum_{k=1}^{K} w_k\hat{B}_k(z) = \hat{\boldsymbol{B}}(z)\,\boldsymbol{w} \quad (43)$$

where $\boldsymbol{w} = [w_1, w_2, \ldots, w_K]^T$ is unknown because the optimal projection function $\omega(z)$ needs to be searched. Next, we build the following optimization model NDA for achieving aggregation and separation of samples by the projection function $\omega(z)$ in the complex function space

$$\begin{aligned} \max_{\boldsymbol{w}} J_c(\boldsymbol{w}) &= \frac{\frac{1}{4\pi^2}\int_0^{2\pi}\int_0^{2\pi}\omega(z)^* F_b(z,\zeta)\omega(\zeta)dzd\zeta}{\frac{1}{4\pi^2}\int_0^{2\pi}\int_0^{2\pi}\omega(z)^* F_t(z,\zeta)\omega(\zeta)dzd\zeta} \\ &= \frac{\boldsymbol{w}^*\boldsymbol{S}_b^c\boldsymbol{w}}{\boldsymbol{w}^*\boldsymbol{S}_t^c\boldsymbol{w}}. \end{aligned} \quad (44)$$

The conclusion in (44) is because of the unit orthogonality of the optimal TM system $\{\hat{B}_k(z)\}_{k=1}^{K}$ in (8) and (9), i.e.,

$$\frac{1}{2\pi}\int_0^{2\pi}\hat{\boldsymbol{B}}(z)^*\,\hat{\boldsymbol{B}}(z)\,dz = I_{K\times K}. \quad (45)$$

Here $I_{K\times K}$ is a $K \times K$ identity matrix. Through the above derivation, we find that the optimization model NDA established for the spectral curves in the complex function space is finally simplified into a discriminant model for their coefficients. Finding the optimal projection function $\hat{\omega}(z)$ also translates into finding the best projection vector $\hat{\boldsymbol{w}}$ for the representation coefficients. As in LDA, the final model in (46) can be solved by solving the following eigenvalue problem:

$$\boldsymbol{S}_b^c\boldsymbol{w} = \lambda\boldsymbol{S}_t^c\boldsymbol{w}. \quad (46)$$

Based on the first $C-1$ optimal projection vectors $\{\hat{\boldsymbol{w}}_j\}_{j=1}^{C-1}$, the optimal projection functions $\{\hat{\omega}_j(z)\}_{j=1}^{C-1}$ can be denoted as follows:

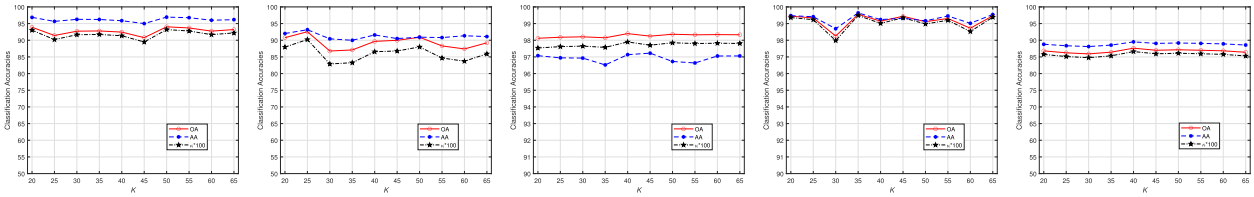
$$\hat{\omega}_j(z) = \hat{\boldsymbol{B}}(z)\,\hat{\boldsymbol{w}}_j, \quad j = 1, 2, \ldots, C-1. \quad (47)$$

Finally, the discriminant feature $\boldsymbol{f}_i = [\boldsymbol{f}_{i1}, \boldsymbol{f}_{i2}, \ldots, \boldsymbol{f}_{i,C-1}]$ of the spectral curve $\hat{f}_i(z)$ can be obtained by each projection function $\hat{\omega}_j(z)$ in the complex function space

$$\boldsymbol{f}_{ij} = \left\langle\hat{f}_i(z), \hat{\omega}_j(z)\right\rangle = \left\langle\hat{\boldsymbol{B}}(z)\,\hat{\boldsymbol{c}}_i, \hat{\boldsymbol{B}}(z)\,\hat{\boldsymbol{w}}_j\right\rangle = \hat{\boldsymbol{c}}_i\hat{\boldsymbol{w}}_j^*. \quad (48)$$

Based on the discriminant features, we can effectively implement the classification tasks of HSIs by SVM. The algorithmic procedure of the NDA is given in Algorithm 2.

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(a) (b) (c) (d) (e)

Fig. 2. Classification accuracies for ORF-NDA with different values of K on five HSI datasets. (a) IP. (b) PU. (c) LK. (d) CH. (e) HoustonU2013.

Algorithm 2 NDA in Complex Function Space

Input: The training set $\{\hat{\boldsymbol{c}}_i^j, y_i^j\}_{i=1}^{n_j}$ derived from spectral curves $\{\hat{f}_i^j, y_i^j\}_{i=1}^{n_j}$, $j = 1, 2, \cdots, C$, all samples $\{\hat{\boldsymbol{c}}_i, y_i\}_{i=1}^{n}$ derived from spectral curves $\{\hat{f}_i, y_i\}_{i=1}^{n}$.

Output: The discriminant features $\{\boldsymbol{f}_i\}_{i=1}^{n}$.

1: Calculate the global center $\boldsymbol{u}_c$ and the center for each class $\boldsymbol{u}_c^j$, $j = 1, 2, \cdots, C$.
2: Calculate the total scatter matrix $\boldsymbol{S}_t^c$.
3: Calculate the between-class scatter matrix $\boldsymbol{S}_b^c$.
4: Solve the eigenvalue problem in (46) and find the first $C-1$ optimal projection vectors $\{\hat{\boldsymbol{w}}_j\}_{j=1}^{C-1}$.
5: Extract the discriminant features $\{\boldsymbol{f}_i\}_{i=1}^{n}$ for all samples based on (48).
6: **Return** $\{\boldsymbol{f}_i\}_{i=1}^{n}$.

D. Framework of HSI Classification Based on NDA

Here, we design a novel spectral–spatial classification model based on the proposed NDA from the perspective of taking a function as the basic processing unit in the complex function space, termed ORF-NDA. The proposed ORF-NDA fully considers the temporal characteristics of the spectral curve using a functional data fitting model based on the optimal ORF system. The ORF system is ingeniously designed using the SAFD method. It can adaptively select the optimal parameters to construct the ORF system according to the changing characteristics of different spectral curves in the HSI. Next, based on the resulting spectral functions, an NDA method is proposed to extract the discriminant features of spectral functions by seeking the optimal projection functions. Due to the properties of the ORF system, the optimization model that is established based on the functional discrimination criterion in the complex space is transformed into an optimization model in the common vector space, and then into a simple eigenvalue solution problem. In the proposed ORF-NDA, in order to utilize the spatial information, we first processed each pixel using a simple mean filter to roughly eliminate noise interference in the spatial dimension before feature extraction using NDA. Finally, the resulting discriminant features are the inputs of the SVM for classification.

III. Experimental Results and Analysis

In this section, we compare the proposed ORF-NDA with several classification methods on five common HSI datasets, including the Indian Pines (IP) scene, Pavia University (PU) scene, WHU-Hi-LongKou (LK) scene, Hyperspectral data for the University of Houston campus and its neighboring area in the 2013 IEEE GRSS Data Fusion Contest (HoustonU2013) [33], and Chikusei (CH) scene [34]; see Table I for details. To demonstrate the effectiveness and advantages of the proposed ORF-NDA, especially the ability to extract and utilize the sequence attributes of spectral data, three benchmark machine learning algorithms (SVM, PCA, and LDA), seven deep learning algorithms (ViT, SpFormer [15], GSC-ViT [35], SSFTT [16], DSNet [11], MAGS [17], and SimHSIC [18]), and two functional data algorithms (FPCA [22] and FDDA [36]) were used for comparison. These comparison methods encompass a wide range of machine learning algorithms, spanning from shallow to deep methods, as well as from traditional discrete multivariate approaches to continuous function-based methods. The comparative analysis of various data processing modes underscores the distinct advantages that the proposed method offers for processing HSI data. These experiments demonstrate the effectiveness of the proposed ORF-NDA in handling and enhancing high-dimensional continuous data, making it a superior choice for HSI data processing tasks.

A. Experimental Design and Parameter Selection

In this article, we focus on the use of spectral information for HSI classification. Meanwhile, in order to fully utilize the spatial information of HSIs while preserving the functional characteristics of the spectral curve, we use local mean filtering to extract the spatial information of HSIs. The neighborhood size for the local mean filtering is uniformly set to 7×7, consistent with the patch size of comparison methods in the article. Except for the patch size, the other parameter settings of all comparison methods were set to be consistent with their References, and their experimental results were obtained based on their shared codes. The benchmark methods obtain their best accuracy through cross-validation. For the proposed ORF-NDA, the most important parameter K is tuned in the interval $[20, 65]$ with an interval of 5. The experiments are implemented using MATLAB R2024a in a PC with Intel(R) Core(TM) i9-14900K CPU and 64G DDR5 RAM.

For the IP, PU, LK, and CH datasets, we select 50 training samples per class (15 samples are selected as training samples for three small classes on the IP dataset), and the remaining samples are used for testing, as shown in Table I. For the HoustonU2013 dataset, the training and test samples have been given in advance, as described in the 2013 IEEE GRSS Data Fusion Contest [33]. It is noteworthy that a large cloud shadow was present during the acquisition of the HoustonU2013 dataset. As a result, no training samples were selected in this

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TABLE I
DESCRIPTION OF DATASETS (TR: NUMBER OF TRAINING SAMPLES, TE: NUMBER OF TEST SAMPLES)

HSIs	Indian Pines (IP)			Pavia University (PU)			WHU-Hi-LongKou (LK)			Chikusei (CH)			HoustonU2013		
Size	$145 \times 145 \times 200$			$610 \times 340 \times 103$			$550 \times 400 \times 270$			$2517 \times 2335 \times 128$			$349 \times 1905 \times 144$		
Class	Name	TR	TE	Name	TR	TE	Name	TR	TE	Name	TR	TE	Name	TR	TE
1	Alfalfa	15	31	Asphalt	50	6581	Corn	50	34461	Water	50	2795	Healthy grass	198	1053
2	Corn-notill	50	1378	Meadows	50	18599	Cotton	50	8324	Bare soil (school)	50	2809	Stressed grass	190	1064
3	Corn-mintill	50	780	Gravel	50	2049	Sesame	50	2981	Bare soil (park)	50	236	Synthetic grass	192	505
4	Corn	50	187	Trees	50	3014	Broad-leaf soybean	50	63162	Bare soil (farmland)	50	4802	Trees	188	1056
5	Grass-pasture	50	433	Painted metal sheets	50	1295	Narrow-leaf soybean	50	4101	Natural plants	50	4247	Soil	186	1056
6	Grass-trees	50	680	Bare Soil	50	4979	Rice	50	11804	Weeds in farmland	50	1058	Water	182	143
7	Grass-pasture-mowed	15	13	Bitumen	50	1280	Water	50	67006	Forest	50	20466	Residential	196	1072
8	Hay-windrowed	50	428	Self-Blocking Bricks	50	3632	Roads and houses	50	7074	Grass	50	6465	Commercial	191	1053
9	Oats	15	5	Shadows	50	897	Mixed weed	50	5179	Rice field (grown)	50	13319	Road	193	1059
10	Soybean-notill	50	922							Rice field (first stage)	50	1218	Highway	191	1036
11	Soybean-mintill	50	2405							Row crops	50	5911	Railway	181	1054
12	Soybean-clean	50	543							Plastic house	50	2143	Parking Lot 1	192	1041
13	Wheat	50	155							Manmade (non-dark)	50	1170	Parking Lot 2	184	285
14	Woods	50	1215							Manmade (dark)	50	7614	Tennis Court	181	247
15	Buildings-Grass-Trees-Drives	50	336							Manmade (blue)	50	381	Running Track	187	473
16	Stone-Steel-Towers	50	43							Manmade (red)	50	172			
17										Manmade grass	50	990			
18										Asphalt	50	751			
19										Paved ground	50	95			

region. However, a significant number of validation samples were collected to test the efficacy of various algorithms in dealing with cloud shadow. Therefore, this scenario is a very challenging task. For every case except the HoustonU2013 dataset, 10 independent runs are performed, and the average results, including the per-class accuracy (PA [%]), overall accuracy (OA [%]), average accuracy (AA [%]), Kappa statistic (κ), and their standard deviations, are reported.

In order to discuss the impact of this important parameter K in the proposed ORF-NDA on the classification results, we conduct the experiments on five HSI datasets and display the accuracy change curve (including OA, AA, and κ) of the proposed ORF-NDA at different values of K in the interval $[20, 65]$ with an interval of 5, as shown in Fig. 2. For the IP dataset, we observed that variations in the parameter K have minimal impact on the classification accuracy of the ORF-NDA method. Consequently, we selected $K = 50$ for subsequent tests. In contrast, accuracy changes were more pronounced for the PU and CH datasets when K was relatively small. However, the accuracy gradually decreases once K exceeded 40 for PU and 35 for CH, leading us to choose $K = 40$ for PU and $K = 35$ for CH. Regarding the LK and HoustonU2013 datasets, while the classification accuracy remained relatively stable with different values of K, it showed a slow decline when K surpassed 40. This decline in accuracy with increasing K is attributed to higher data dimensionality, which, although it reduces fitting errors, also diminishes the system's noise resistance capability. Therefore, we opt for $K = 40$ for the LK and HoustonU2013 datasets.

Fig. 3 illustrates the reconstruction performance of the optimal ORF system derived by SAFD within the ORF-NDA framework on the PU dataset with $K = 40$. It displays the reconstructed spectral vector (RS) obtained by sampling from the fitted curve of each land cover, and the corresponding absolute error (AE) between the original spectral vector (OS) and the RS from Class 1 to Class 9. The AEs between the OS and RS in Classes 1, 3, and 9 are notably larger than in other classes. Furthermore, the influence of K in the proposed method ORF-NDA on the function data fitting effect is discussed, as shown in Fig. 4. The mean square error (mse) between OS and RS from Class 1 to Class 9 under different

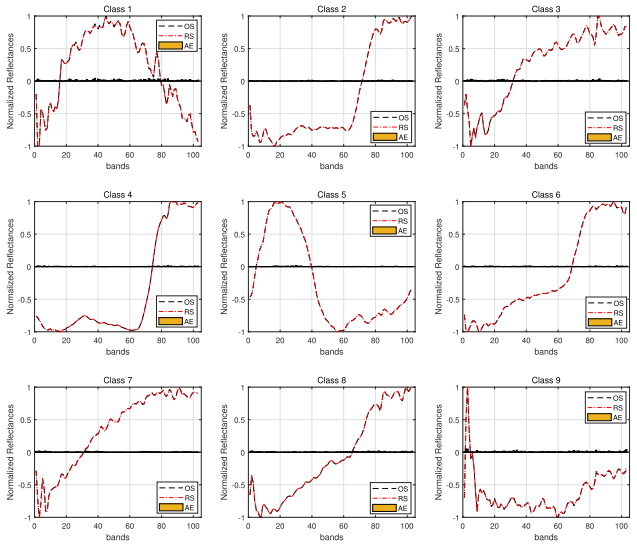


Fig. 3. Functional data fitting effect of the proposed ORF-NDA for different land covers from Class 1 to Class 9 on the PU dataset. Here, OS denotes the OS, RS is the RS obtained by sampling from the corresponding fitted curve, and AE signifies the absolute error between the OS and RS.

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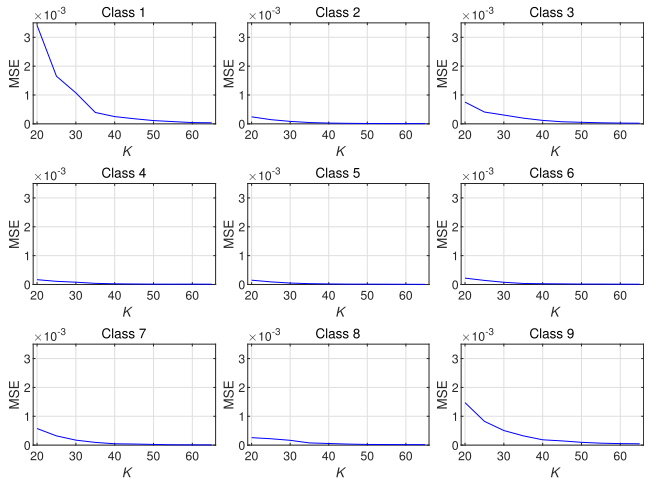


Fig. 4. MSE of the OS and RS from Class 1 to Class 9 with different values of K on the PU dataset.

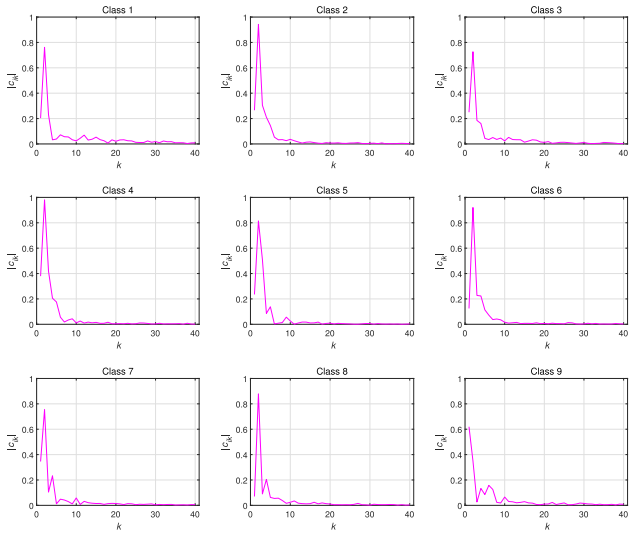
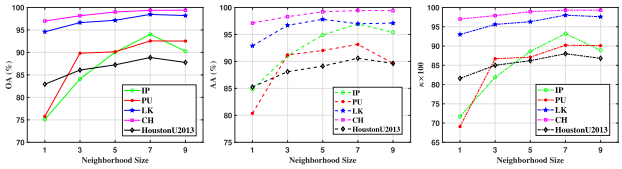


Fig. 5. Amplitude change of the representation coefficients for the spectral curves across Class 1 to Class 9 on the PU dataset with k varying from the initial term to the final term when $K = 40$.



(a) (b) (c)

Fig. 6. ORF-NDA performance under local mean filtering with varying neighborhood sizes on five HSI datasets. (a) OA. (b) AA. (c) κ. Neighborhood size = 1 indicates no mean filtering (original spectral information).

values of K is displayed on the PU datasets. It is clearly shown that as K increases, the fitting effect becomes better and the mse becomes lower. Notably, when K reaches 40, the MSEs for all classes except Classes 1, 3, and 9 converge to near-zero values, consistent with the trend observed in Fig. 3. In particular, we discuss the variation of the representation coefficient of the spectral curve. Due to the orthogonality and normalization of the optimal TM system $\{\hat{B}_k(z)\}_{k=1}^K$, the sum of the energy, $\sum_{k=1}^K |\hat{c}_{ik}|^2$, of the representation coefficients, $\{\hat{c}_{ik}\}_{k=1}^K$, is approximately equal to the entire energy, $\|f_i\|_2^2$, of the ith spectral curve, as described in (36). Based on the stepwise optimization model (19), (27) and the ensuing recursion formula, the energy, $\{|\hat{c}_{ik}|^2\}_{k=1}^K$, of the representation coefficients, $\{\hat{c}_{ik}\}_{k=1}^K$, exhibits a decreasing trend, with the exception of the initial term which represents the average value of the spectral curve. Fig. 5 illustrates the amplitude change of the representation coefficients for spectral curves spanning Classes 1 to 9 on the PU dataset as k progresses from the first to the last term when $K = 40$. The results presented in Fig. 5 demonstrate that, apart from the initial term, the energy, $\{|\hat{c}_{ik}|^2\}_{k=1}^K$, of the representation coefficients is also decreasing as k increases. This finding confirms the validity of the proposed methodology that integrates the energy minimization principle for residuals into parameter optimization processes. It also validates the efficacy of the proposed model (17) in achieving high-precision fitting outcomes while employing a sparse set of basis functions.

Fig. 6 illustrates the performance of ORF-NDA under local mean filtering with varying neighborhood sizes across the five HSI datasets. As the neighborhood size increases from 1 to 9, the classification accuracy generally improves in the beginning due to enhanced spatial smoothing, which reduces noise and strengthens spatial coherence. For most datasets, peak performance is observed at the moderate neighborhood size 7×7, indicating an optimal balance between noise suppression and preservation of fine spatial details. Beyond this range, performance tends to plateau or slightly decline, likely due to excessive smoothing that blurs class boundaries and weakens discriminative features. These results demonstrate that spatial preprocessing via local mean filtering significantly enhances the classification capability of ORF-NDA, with optimal gains achieved at intermediate neighborhood sizes.

B. Experimental Results and Comparison

To validate the superiority of the proposed method, we evaluated its classification performance on the IP dataset and compared it against several benchmark and state-of-the-art algorithms. The experimental results are shown in Table II. From the perspective of accuracy of pure spectral classification (without considering spatial information), the functional data learning method, FPCA, is significantly superior to ViT and three machine learning algorithms (SVM, PCA, and LDA) under the given number of training samples in Table I. This reflects the advantages of functional data learning algorithms in processing high-dimensional continuous spectral data in HSIs with few training samples. When training samples are limited, ViT with pixelwise input performs even worse in terms of classification accuracy than SVM. This indicates that for high-dimensional spectral data (e.g., 200 bands), ViT's ability to process and extract meaningful features is inferior to SVM under data-scarce conditions. Although deep learning models, like ViT, have great potential, they require massive amounts of data and very rigorous training to achieve optimal results. Furthermore, in the case of limited training samples,

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TABLE II

CLASSIFICATION ACCURACIES ON THE IP DATASET OBTAINED BY DIFFERENT METHODS. THE BEST ACCURACIES ARE SHOWN IN BOLD. ALL VALUES ARE IN PERCENTAGE EXCEPT FOR κ

Class	Conventional Algorithms			Deep Learning Algorithms							Functional Data Learning Algorithms		
	SVM	**PCA**	**LDA**	**ViT**	**SpFormer**	**GSC-ViT**	**SSFTT**	**DSNet**	**MAGS**	**SimHSIC**	**FPCA**	**FDDA**	**ORF-NDA**
1	78.71 ± 2.18	80.00 ± 2.72	75.48 ± 1.56	79.49 ± 12.68	79.49 ± 2.53	99.57 ± 1.37	94.87 ± 3.07	94.87 ± 1.84	98.91 ± 2.23	96.55 ± 2.61	85.81 ± 2.74	98.71 ± 3.12	**99.68** ± 1.02
2	58.31 ± 4.31	54.57 ± 4.14	61.31 ± 6.97	53.25 ± 6.81	70.52 ± 0.01	82.73 ± 4.96	85.62 ± 4.05	83.96 ± 3.38	77.68 ± 13.72	80.69 ± 5.36	67.34 ± 2.95	90.17 ± 3.55	**92.32** ± 3.22
3	69.35 ± 3.17	67.21 ± 2.79	60.47 ± 2.20	66.20 ± 10.12	81.89 ± 3.98	91.33 ± 4.01	90.43 ± 3.14	**94.90** ± 2.72	92.22 ± 5.45	82.49 ± 7.59	71.79 ± 3.24	92.14 ± 1.88	93.35 ± 1.72
4	77.06 ± 0.80	77.27 ± 0.81	80.75 ± 1.32	86.41 ± 4.91	91.30 ± 0.39	98.78 ± 1.17	97.28 ± 1.10	99.46 ± 0.58	99.28 ± 1.05	97.90 ± 1.70	81.50 ± 1.77	98.66 ± 2.20	**100.00** ± 0.00
5	90.62 ± 3.19	90.88 ± 3.34	90.42 ± 3.50	89.71 ± 9.14	95.53 ± 3.82	93.31 ± 5.30	90.60 ± 1.40	**98.21** ± 2.42	95.28 ± 3.17	91.94 ± 2.84	91.94 ± 3.02	95.50 ± 3.10	94.90 ± 2.55
6	93.62 ± 0.88	92.56 ± 0.82	92.00 ± 0.86	87.66 ± 2.88	85.51 ± 2.17	98.70 ± 1.48	95.95 ± 0.56	97.27 ± 1.04	**99.79** ± 0.36	98.25 ± 0.85	92.85 ± 0.91	98.56 ± 0.54	98.16 ± 1.79
7	85.38 ± 2.43	87.69 ± 2.43	91.54 ± 0.00	90.91 ± 5.06	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	98.93 ± 3.21	**100.00** ± 0.00	86.92 ± 2.43	**100.00** ± 0.00	**100.00** ± 0.00
8	95.37 ± 1.18	95.23 ± 1.18	96.68 ± 0.00	89.98 ± 2.37	99.32 ± 0.69	99.96 ± 0.13	98.63 ± 0.21	99.54 ± 0.52	99.81 ± 0.27	99.82 ± 0.41	94.81 ± 1.82	**100.00** ± 0.00	99.98 ± 0.07
9	86.00 ± 0.00	82.00 ± 0.00	94.00 ± 0.00	80.00 ± 2.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	96.00 ± 8.43	**100.00** ± 0.00	**100.00** ± 0.00
10	73.36 ± 3.84	71.31 ± 4.64	70.21 ± 2.17	72.22 ± 5.16	81.81 ± 1.48	87.33 ± 5.40	85.51 ± 3.12	**95.10** ± 2.88	80.77 ± 9.88	91.63 ± 4.85	76.44 ± 3.84	87.15 ± 2.91	92.51 ± 2.75
11	56.75 ± 2.97	55.65 ± 3.93	58.67 ± 4.79	66.00 ± 7.82	75.48 ± 5.53	81.49 ± 4.74	75.27 ± 4.95	**90.07** ± 6.82	78.52 ± 5.02	84.87 ± 3.32	59.28 ± 3.70	83.87 ± 3.26	89.45 ± 2.36
12	72.06 ± 6.07	68.23 ± 5.96	72.74 ± 3.03	57.09 ± 6.74	73.76 ± 3.35	93.00 ± 3.93	78.90 ± 3.95	**94.68** ± 1.70	87.37 ± 5.51	92.79 ± 3.04	78.01 ± 4.41	89.15 ± 2.75	91.82 ± 2.23
13	98.45 ± 1.20	97.74 ± 1.22	98.45 ± 0.62	97.53 ± 0.49	98.77 ± 0.71	**100.00** ± 0.00	**100.00** ± 0.00	99.38 ± 0.32	99.66 ± 0.38	99.89 ± 0.23	98.71 ± 1.10	99.74 ± 0.45	99.81 ± 0.31
14	88.44 ± 1.67	88.75 ± 1.63	84.57 ± 0.76	87.62 ± 13.68	93.17 ± 1.76	95.16 ± 3.39	94.21 ± 2.24	95.90 ± 1.88	96.72 ± 2.32	94.48 ± 2.32	89.40 ± 1.57	95.18 ± 2.57	**99.16** ± 0.58
15	62.47 ± 1.91	59.17 ± 1.94	70.45 ± 0.56	63.94 ± 7.73	78.48 ± 8.64	97.10 ± 2.72	92.73 ± 1.94	97.58 ± 0.93	98.55 ± 2.40	92.91 ± 3.54	64.05 ± 1.91	98.24 ± 1.09	**99.91** ± 0.20
16	95.58 ± 3.06	95.35 ± 2.14	93.49 ± 1.64	95.56 ± 0.32	**100.00** ± 0.00	99.46 ± 0.76	97.78 ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	96.28 ± 2.64	98.60 ± 1.63	**100.00** ± 0.00
OA	71.97 ± 0.79	70.42 ± 0.91	71.68 ± 1.62	71.86 ± 2.36	81.76 ± 1.74	89.70 ± 1.19	86.51 ± 1.11	93.08 ± 1.92	87.82 ± 3.39	89.34 ± 1.34	75.03 ± 0.81	91.00 ± 1.04	**94.03** ± 0.99
AA	80.10 ± 0.49	78.98 ± 0.40	80.70 ± 0.73	78.97 ± 1.40	87.81 ± 0.20	94.87 ± 0.51	92.33 ± 0.42	96.31 ± 0.62	93.97 ± 1.79	94.01 ± 0.78	83.20 ± 0.83	95.35 ± 0.49	**96.94** ± 0.42
κ	0.683 ± 0.009	0.666 ± 0.010	0.680 ± 0.018	0.680 ± 0.025	0.792 ± 0.019	0.883 ± 0.013	0.846 ± 0.012	0.921 ± 0.022	0.862 ± 0.038	0.878 ± 0.015	0.718 ± 0.009	0.897 ± 0.012	**0.932** ± 0.011

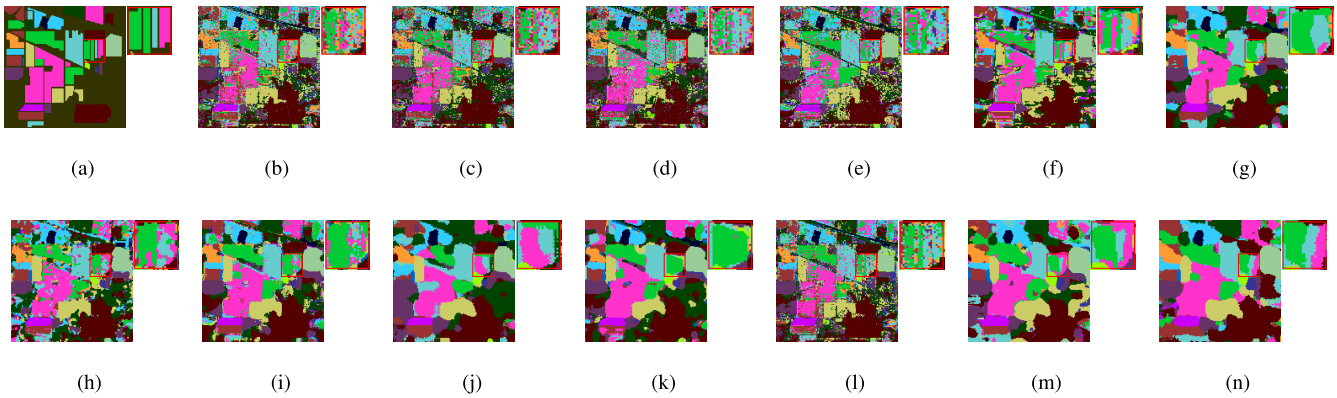


Fig. 7. Classification maps for different methods on the IP dataset. In each subfigure, a zoomed-in subregion has been added on the top-right to highlight performance on local structures. (a) Ground truth. (b) SVM. (c) PCA. (d) LDA. (e) ViT. (f) SpFormer. (g) GSC-ViT. (h) SSFTT. (i) DSNet. (j) MAGS. (k) SimHSIC. (l) FPCA. (m) FDDA. (n) ORF-NDA.

the feature extraction effects of PCA and LDA have no advantage over directly using raw spectral data for classification. After applying techniques that incorporate spatial information (including SpFormer with patchwise input, GSC-ViT featuring patchwise input, SSFTT combining 3-D convolution and 2-D convolution kernels, DSNet integrating 2D-CNN network, MAGS with spatial patch input, SimHSIC featuring 3-D spatial–spectral patch, FDDA enhanced by mean filtering, and ORF-NDA utilizing mean filtering), it is observed that spectral–spatial methodologies demonstrate superior classification accuracies when compared to the three aforementioned machine learning approaches and ViT with pixelwise input. The spectral–spatial methods integrate spatial and spectral data effectively, leading to more accurate and reliable classification results in complex image analysis tasks. In particular, the spectral–spatial functional learning methods FDDA and ORF-NDA outperform five advanced deep learning methods (SpFormer, GSC-ViT, SSFTT, MAGS, and SimHSIC) under limited training samples, which demonstrates the effectiveness of functional learning methods and the stronger capability to capture long-range dependencies and nonlinear interaction relationships within high-dimensional continuous spectral data. The proposed ORF-NDA achieves the best classification accuracies, which are much higher than other comparison methods, including FDDA and DSNet. This illustrates the superiority of the functional data fitting model using SAFD and the NDA method in complex function space. They can ensure better basis functions to achieve effective fitting of spectral curves and extract more discriminative features of spectral curves.

In order to more intuitively demonstrate the advantages of the proposed method, we provide the classification maps of all methods mentioned above for comparison, and the results are shown in Fig. 7. For all spectral classification methods encompassing three machine learning algorithms (SVM, PCA, and LDA), a benchmark deep learning approach (ViT), and one functional data learning method (FPCA), the classification maps uniformly exhibit varying degrees of "salt and pepper" noise due to their common limitation in utilizing spatial information. There is a large number of misclassifications within some classes. In contrast, the spectral–spatial classification methods produce smoother classification maps, especially in Fig. 7(g), (i), (j), (k), (m), and (n). However, these methods also exhibit varying degrees of class boundary degradation, for

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TABLE III

CLASSIFICATION ACCURACIES ON THE PU DATASET OBTAINED BY DIFFERENT METHODS. THE BEST ACCURACIES ARE SHOWN IN BOLD. ALL VALUES ARE IN PERCENTAGE EXCEPT FOR κ

Class	Conventional Algorithms			Deep Learning Algorithms							Functional Data Learning Algorithms		
	SVM	**PCA**	**LDA**	**ViT**	**SpFormer**	**GSC-ViT**	**SSFTT**	**DSNet**	**MAGS**	**SimHSIC**	**FPCA**	**FDDA**	**ORF-NDA**
1	73.26 ± 2.43	73.85 ± 3.29	56.84 ± 4.66	59.88 ± 3.41	71.40 ± 3.37	**93.95** ± 2.61	83.92 ± 1.31	86.67 ± 7.92	85.18 ± 5.50	91.12 ± 2.36	74.37 ± 4.05	79.61 ± 5.28	86.26 ± 4.25
2	70.40 ± 4.12	72.43 ± 6.18	69.74 ± 6.54	30.68 ± 8.63	82.67 ± 0.31	88.61 ± 5.04	**96.21** ± 2.11	94.34 ± 5.60	92.79 ± 6.57	87.63 ± 5.23	74.74 ± 2.80	91.35 ± 4.93	93.35 ± 1.75
3	72.21 ± 3.85	71.42 ± 3.30	68.81 ± 3.20	55.00 ± 14.77	83.80 ± 1.55	94.71 ± 2.10	84.33 ± 5.97	86.45 ± 18.55	79.85 ± 14.48	90.26 ± 3.94	71.30 ± 2.46	88.13 ± 4.30	**97.76** ± 1.24
4	94.86 ± 1.52	94.12 ± 2.21	94.48 ± 1.42	93.60 ± 2.19	92.87 ± 1.31	95.12 ± 4.28	**98.79** ± 0.50	96.82 ± 0.73	91.46 ± 13.20	92.70 ± 2.51	92.77 ± 1.44	89.07 ± 3.79	93.20 ± 1.21
5	98.29 ± 0.65	98.11 ± 1.10	99.85 ± 0.07	99.31 ± 0.13	**100.00** ± 0.00	99.79 ± 0.26	99.85 ± 0.09	**100.00** ± 0.00	99.97 ± 0.09	**100.00** ± 0.00	99.31 ± 0.56	98.66 ± 1.18	**100.00** ± 0.00
6	70.19 ± 3.25	69.26 ± 5.51	67.74 ± 6.25	91.97 ± 6.29	74.01 ± 8.31	97.23 ± 3.12	64.13 ± 15.42	77.07 ± 18.89	93.78 ± 11.03	91.46 ± 7.00	63.63 ± 5.37	96.90 ± 2.59	**99.92** ± 0.03
7	81.80 ± 3.14	81.06 ± 3.41	70.15 ± 4.99	94.30 ± 1.88	91.41 ± 1.44	98.97 ± 1.14	99.55 ± 0.77	94.81 ± 5.37	98.17 ± 1.94	99.22 ± 1.29	87.19 ± 5.29	99.48 ± 0.54	**99.84** ± 0.06
8	73.17 ± 4.57	71.54 ± 4.35	60.73 ± 6.11	68.72 ± 10.67	85.68 ± 6.44	**93.28** ± 4.33	89.46 ± 1.59	73.84 ± 20.81	87.94 ± 4.16	91.17 ± 3.19	68.17 ± 4.97	85.50 ± 5.06	83.04 ± 2.81
9	97.58 ± 1.09	95.84 ± 5.54	94.93 ± 2.68	**99.89** ± 0.05	**99.89** ± 0.22	99.47 ± 0.46	**99.89** ± 0.08	98.83 ± 0.49	98.22 ± 1.23	99.21 ± 0.20	97.99 ± 0.89	86.47 ± 2.74	85.06 ± 1.78
OA	74.66 ± 1.53	75.24 ± 2.17	69.91 ± 2.84	56.84 ± 3.14	82.10 ± 0.75	92.53 ± 2.11	89.86 ± 1.18	89.44 ± 3.19	91.09 ± 3.31	90.40 ± 1.90	75.55 ± 1.48	89.73 ± 2.40	**92.56** ± 1.17
AA	81.31 ± 0.64	80.85 ± 1.12	75.92 ± 1.26	77.04 ± 0.38	86.86 ± 1.10	**95.68** ± 0.59	90.68 ± 1.05	89.87 ± 3.05	91.93 ± 3.25	93.64 ± 0.74	81.05 ± 0.89	90.58 ± 0.86	93.16 ± 0.82
κ	0.679 ± 0.017	0.685 ± 0.024	0.620 ± 0.031	0.496 ± 0.029	0.768 ± 0.011	**0.903** ± 0.026	0.865 ± 0.015	0.860 ± 0.041	0.884 ± 0.043	0.875 ± 0.024	0.686 ± 0.018	0.866 ± 0.030	0.902 ± 0.015

example, in the classification maps of SpFormer [Fig. 7(f)] and SSFTT [Fig. 7(h)]. The proposed ORF-NDA yields the best-performing classification map [Fig. 7(n)], though it also shows some blurring of class boundaries. However, although DSNet, GSC-ViT, SimHSIC, MAGS, and FDDA yield notably smooth classification maps, these methods exhibit severe misclassifications in several classes with a small number of samples, leading to the misclassification of entire patches, as shown in the zoomed-in regions.

Compared with the IP HSI dataset, the number of spectral bands in the PU HSI dataset is much smaller, and the distribution of land covers is more complex in that scene. In order to further verify the advantages of our proposed method, we verified the proposed method on the PU HSI dataset with a different number of bands and a different distribution of land covers. The quantitative experimental results are shown in Table III. ORF-NDA and GSC-ViT are significantly better than other comparison methods in terms of classification accuracy under the small sample case. The proposed ORF-NDA obtains the highest OA, while GSC-ViT obtains the highest AA and κ, which reflects the overall performance advantage of the proposed ORF-NDA, and the advantages of GSC-ViT in the balance and consistency of classification accuracies for each class. This can also be confirmed by the classification accuracies of land covers in each class. It also confirms the significant advantages of the proposed method in terms of spectral data processing capabilities. The state-of-the-art GSC-ViT can achieve such good classification results on the one hand, because the deep framework more fully mines and utilizes the spatial information of the PU HSI by a groupwise separable convolution (GSC) module, on the other hand, since the PU HSI dataset has fewer spectral bands. The classification performance of different spectral methods exhibits significant divergence in the PU HSI dataset. Notably, ViT's performance falls markedly below expectations, yielding results inferior to conventional classification algorithms. Compared with PCA and LDA, FPCA and FDDA reflect the advantages of the functional data learning models. In addition, comparing ViT with SpFormer, it was found that the use of spatial information significantly improves the classification performance of various methods on the PU HSI dataset. MAGS, SimHSIC, SSFTT, and DSNet also achieve relatively good classification results due to the utilization of spatial information by the patching technique.

Fig. 8 presents the classification maps of the PU scene for all experimental methods, including three conventional algorithms, seven deep learning algorithms, and three functional data learning algorithms as listed in Table III. ORF-NDA, GSC-ViT, MAGS, and SimHSIC all produce relatively smooth classification maps, corresponding to Fig. 8(g), (j), (k), and (n), respectively. A detailed comparison with the ground truth map [Fig. 8(a)] indicates that the classifications derived from MAGS and SimHSIC involve a certain degree of misclassification for specific classes. In contrast, although the classification maps of ORF-NDA and GSC-ViT suffer from a slight loss of edge information, the class distributions are highly consistent with the ground truth. Compared with SpFormer, FDDA achieves a relatively smooth classification map, as depicted in Fig. 8(m), despite less accurate results in some small classes. The classification maps Fig. 8(h) and (i) generated by SSFTT and DSNet preserve some class boundaries effectively, albeit with a notable number of misclassifications in certain blocks. In contrast, other comparative methods that do not utilize spatial information display various degrees of salt and pepper noise in their classification maps, as shown in the zoomed-in regions.

In order to better demonstrate the effectiveness of the proposed method on a high-resolution HSI dataset that includes a complex vegetation scene, the proposed ORF-NDA and the comparison methods were applied on the LK dataset. Table IV presents the classification accuracies for individual classes, OA, AA, and κ for various methods on the LK dataset. The proposed ORF-NDA obtains the best OA of 98.46% and κ of 0.980, which again demonstrates the overall performance advantage of the proposed ORF-NDA. The GSC-ViT also produces very competitive classification accuracies. This illustrates the benefit of this method in fusing local and global spatial features. Due to the high spatial resolution and spatial consistency of the land cover classes in the LK dataset, mining spatial features can effectively enhance the classification accuracy. SSFTT, MAGS, SimHSIC, and FDDA also verified this in our experiments. Although DSNet also uses small-scale convolution, its ability to capture spatial features is slightly worse. However, it still outperforms SpFormer, ViT,

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TABLE IV

CLASSIFICATION ACCURACIES ON THE LK DATASET OBTAINED BY DIFFERENT METHODS. THE BEST ACCURACIES ARE SHOWN IN BOLD. ALL VALUES ARE IN PERCENTAGE EXCEPT FOR κ

Class	Conventional Algorithms			Deep Learning Algorithms							Functional Data Learning Algorithms		
	SVM	**PCA**	**LDA**	**ViT**	**SpFormer**	**GSC-ViT**	**SSFTT**	**DSNet**	**MAGS**	**SimHSIC**	**FPCA**	**FDDA**	**ORF-NDA**
1	92.43 ± 1.58	91.93 ± 1.42	91.13 ± 2.23	68.39 ± 18.31	89.70 ± 11.01	97.67 ± 1.81	96.79 ± 2.72	98.64 ± 0.34	99.61 ± 0.37	98.78 ± 0.82	93.45 ± 1.63	99.30 ± 0.94	**99.79** ± 0.07
2	83.15 ± 3.54	80.99 ± 3.86	75.86 ± 3.68	63.62 ± 22.00	90.73 ± 8.72	97.49 ± 2.54	96.20 ± 3.61	90.86 ± 8.60	95.13 ± 3.13	**98.02** ± 1.33	87.97 ± 2.25	95.10 ± 1.48	95.75 ± 1.30
3	94.17 ± 1.48	93.53 ± 1.47	97.94 ± 0.56	68.98 ± 19.50	90.96 ± 7.60	97.47 ± 2.97	97.59 ± 1.88	92.40 ± 15.08	99.78 ± 0.39	99.08 ± 0.97	94.88 ± 1.13	**99.81** ± 0.16	99.70 ± 0.05
4	84.40 ± 3.16	82.75 ± 4.74	88.35 ± 1.33	68.49 ± 21.00	92.69 ± 5.07	97.66 ± 2.49	96.72 ± 2.43	86.55 ± 9.14	92.41 ± 2.03	90.35 ± 2.83	86.92 ± 3.79	96.76 ± 0.77	**98.70** ± 0.38
5	89.27 ± 2.27	87.55 ± 3.03	85.26 ± 3.92	68.16 ± 21.15	88.14 ± 9.29	97.72 ± 1.95	96.61 ± 2.65	94.31 ± 6.12	98.72 ± 0.65	98.60 ± 0.84	89.14 ± 2.48	**98.78** ± 0.74	98.72 ± 0.05
6	97.17 ± 1.66	96.95 ± 1.47	98.44 ± 0.67	69.79 ± 21.90	92.23 ± 7.87	97.66 ± 3.29	96.82 ± 2.50	97.48 ± 1.72	98.61 ± 2.98	99.19 ± 0.46	98.47 ± 0.77	97.52 ± 3.17	**99.72** ± 0.00
7	99.57 ± 0.22	99.56 ± 0.21	99.76 ± 0.11	69.97 ± 20.04	91.35 ± 5.50	98.07 ± 2.05	95.99 ± 4.40	99.29 ± 0.56	99.16 ± 0.48	**99.84** ± 0.06	99.79 ± 0.08	99.16 ± 0.77	99.08 ± 0.07
8	86.98 ± 2.16	85.46 ± 2.25	89.55 ± 1.68	66.65 ± 21.97	90.40 ± 5.97	**97.05** ± 3.32	97.00 ± 2.87	93.53 ± 2.51	95.67 ± 2.29	95.64 ± 1.45	86.76 ± 1.67	90.56 ± 2.92	88.18 ± 0.55
9	85.61 ± 2.69	84.12 ± 2.84	87.76 ± 4.44	67.61 ± 23.80	90.96 ± 8.48	**98.43** ± 1.94	96.77 ± 2.29	90.09 ± 4.23	94.28 ± 3.68	87.95 ± 3.20	87.70 ± 1.78	91.19 ± 3.15	93.15 ± 3.06
OA	91.78 ± 0.83	90.95 ± 1.53	92.74 ± 0.73	75.85 ± 3.18	93.96 ± 1.34	97.84 ± 0.42	96.96 ± 0.82	94.16 ± 2.90	96.71 ± 0.64	96.13 ± 0.94	93.14 ± 1.18	97.68 ± 0.59	**98.46** ± 0.08
AA	90.30 ± 0.48	89.21 ± 1.09	90.45 ± 0.80	68.28 ± 2.08	90.65 ± 1.35	**97.69** ± 0.37	96.71 ± 0.43	93.68 ± 2.21	97.04 ± 0.48	96.38 ± 0.62	91.68 ± 0.65	96.46 ± 0.63	96.98 ± 0.23
κ	0.894 ± 0.010	0.88 ± 0.019	0.906 ± 0.009	0.699 ± 0.037	0.922 ± 0.017	0.972 ± 0.005	0.960 ± 0.011	0.925 ± 0.036	0.957 ± 0.008	0.950 ± 0.012	0.911 ± 0.015	0.970 ± 0.008	**0.980** ± 0.001

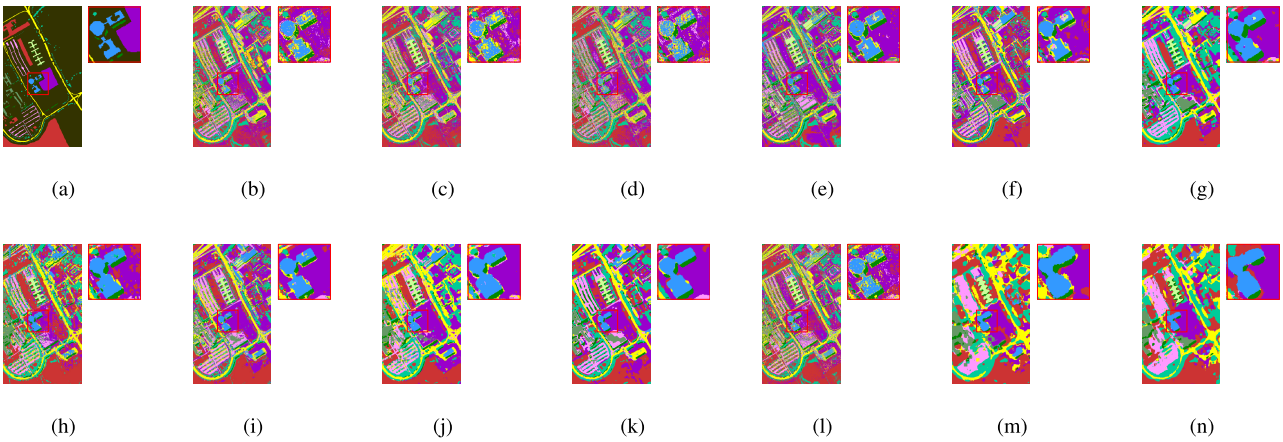


Fig. 8. Classification maps for different methods on the PU dataset. In each subfigure, a zoomed-in subregion has been added on the top-right to highlight performance on local structures. (a) Ground truth. (b) SVM. (c) PCA. (d) LDA. (e) ViT. (f) SpFormer. (g) GSC-ViT. (h) SSFTT. (i) DSNet. (j) MAGS. (k) SimHSIC. (l) FPCA. (m) FDDA. (n) ORF-NDA.

FPCA, and three conventional algorithms. Furthermore, FPCA achieves better classification accuracy than SVM, PCA, LDA, and ViT due to its functional feature extraction capability for high spectral resolution data.

To more intuitively demonstrate the classification effects of various algorithms, including the conventional, deep learning, and functional data learning algorithms in this LK scenario, Fig. 9 shows the classification maps of the corresponding algorithms on the LK HSI dataset, and the zoomed-in regions in the maps. The proposed ORF-NDA still produces the classification map [see Fig. 9(n)], which is closest to the ground truth in Fig. 9(a). In contrast, the classification maps generated by other methods have their classification edges destroyed, and some blocks are also misclassified, as shown in the zoomed-in regions. In addition, the classification maps, Fig. 9(b)–(e), (h), and (l), generated by SVM, PCA, LDA, ViT, SSFTT, and FPCA, have different degrees of salt and pepper noise. Although other methods (like SpFormer, GSC-ViT, DSNet, MAGS, SimHSIC, and FDDA) produce relatively smooth classification maps, some areas and some small classes are misclassified.

To further validate the effectiveness and advantages of the proposed ORF-NDA in a large-area complex scenario, we present the classification results of all methods on the Chikusei (CH) dataset in Table V. This dataset covers a spatial extent of 2517 × 2335 pixels with 128 spectral bands. Highly scattered land covers make it well-suited for evaluating model performance in complex landscapes. ORF-NDA achieves the highest overall performance metrics with an OA of 99.37%, an AA of 99.44%, and a κ of 0.993. This confirms its superior ability to generalize across the large-area complex dataset. Moreover, ORF-NDA excels in per-class accuracy while maintaining stability. It achieves 100% accuracy for four key classes and the highest PA for ten classes. FDDA, GSC-ViT, and DSNet obtained competitive results. This demonstrates the advantages of functional data processing, spectral grouping convolution, and information fusion across pixel and subpixel features in capturing spatial–spectral dependencies within hyperspectral data, respectively. SpFormer, SSFTT, and SimHSIC also achieved good classification accuracies due to their use of the Transformer architecture. In contrast, ViT still underperformed relative to three traditional benchmark methods (SVM, PCA, and LDA) and FPCA in this small-sample classification task. Furthermore, we give the whole classification maps and zoomed-in regions of the CH scene in Fig. 10 for various methods. Given the large-area, highly scattered ground objects in this complex scenario, most methods produce reasonably classified maps [compared to Fig. 10(a)] that roughly capture

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TABLE V

CLASSIFICATION ACCURACIES ON THE CH DATASET OBTAINED BY DIFFERENT METHODS. THE BEST ACCURACIES ARE SHOWN IN BOLD. ALL VALUES ARE IN PERCENTAGE EXCEPT FOR κ

Class	Conventional Algorithms			Deep Learning Algorithms							Functional Data Learning Algorithms		
	SVM	**PCA**	**LDA**	**ViT**	**SpFormer**	**GSC-ViT**	**SSFTT**	**DSNet**	**MAGS**	**SimHSIC**	**FPCA**	**FDDA**	**ORF-NDA**
1	98.62 ± 1.76	98.63 ± 1.77	98.91 ± 1.80	94.23 ± 4.81	97.51 ± 2.38	97.72 ± 2.53	99.88 ± 0.11	97.52 ± 2.58	**99.98** ± 0.04	99.78 ± 0.27	98.70 ± 1.79	98.65 ± 2.14	99.45 ± 0.57
2	98.62 ± 1.20	98.60 ± 1.20	98.36 ± 1.32	95.32 ± 1.46	97.78 ± 0.88	99.55 ± 0.82	97.87 ± 0.79	98.91 ± 1.41	97.39 ± 1.43	99.36 ± 0.45	98.65 ± 1.16	99.54 ± 1.19	**99.97** ± 0.03
3	95.17 ± 2.20	94.49 ± 2.17	95.17 ± 2.39	95.25 ± 3.31	93.39 ± 5.91	99.90 ± 0.16	98.88 ± 1.26	96.99 ± 7.07	99.72 ± 0.54	99.69 ± 0.34	95.13 ± 2.04	**100.00** ± 0.00	**100.00** ± 0.00
4	88.46 ± 3.75	87.85 ± 4.08	91.26 ± 1.36	68.10 ± 6.89	76.85 ± 5.37	96.90 ± 1.58	92.14 ± 2.30	88.28 ± 18.64	87.83 ± 4.51	81.89 ± 5.65	91.96 ± 2.28	97.89 ± 1.28	**98.52** ± 1.09
5	99.12 ± 0.43	99.10 ± 0.41	99.64 ± 0.20	98.31 ± 0.49	98.99 ± 0.88	99.56 ± 0.56	99.97 ± 0.06	99.55 ± 0.41	99.71 ± 0.27	99.96 ± 0.06	99.15 ± 0.38	**100.00** ± 0.00	**100.00** ± 0.00
6	95.87 ± 1.21	95.80 ± 1.14	95.45 ± 0.85	93.71 ± 4.69	98.02 ± 1.94	99.71 ± 0.64	95.52 ± 1.86	99.01 ± 1.67	96.51 ± 1.93	97.88 ± 2.21	95.56 ± 1.83	99.68 ± 0.40	**99.80** ± 0.42
7	92.66 ± 1.39	92.35 ± 1.56	96.01 ± 1.05	94.41 ± 4.89	99.39 ± 0.57	99.62 ± 0.67	98.01 ± 0.74	99.93 ± 0.09	98.73 ± 0.89	**99.96** ± 0.10	93.32 ± 2.14	99.81 ± 0.30	99.72 ± 0.20
8	96.96 ± 0.87	96.76 ± 0.92	96.34 ± 1.28	95.75 ± 1.05	98.72 ± 1.06	98.90 ± 1.54	97.68 ± 1.84	95.11 ± 13.21	97.32 ± 2.42	99.19 ± 1.05	95.82 ± 1.35	99.78 ± 0.07	**99.84** ± 0.06
9	98.55 ± 0.82	98.50 ± 0.90	98.42 ± 0.62	96.03 ± 3.96	98.18 ± 1.62	99.41 ± 0.67	97.21 ± 2.27	99.68 ± 0.26	97.62 ± 2.81	99.24 ± 1.07	98.58 ± 0.72	99.30 ± 0.65	**99.69** ± 0.20
10	99.59 ± 0.39	99.60 ± 0.39	99.83 ± 0.21	99.69 ± 0.35	99.74 ± 0.39	99.70 ± 0.47	99.98 ± 0.03	99.91 ± 0.29	99.85 ± 0.29	**100.00** ± 0.00	99.82 ± 0.22	99.89 ± 0.22	99.85 ± 0.34
11	97.35 ± 1.39	97.30 ± 1.36	96.14 ± 1.24	94.25 ± 5.69	99.00 ± 1.46	99.68 ± 0.47	96.36 ± 1.34	99.14 ± 2.38	95.61 ± 3.87	97.60 ± 3.01	97.32 ± 1.22	**99.73** ± 0.33	99.69 ± 0.30
12	93.15 ± 1.67	93.21 ± 1.69	92.83 ± 1.90	94.27 ± 2.98	98.39 ± 0.96	98.24 ± 1.48	94.40 ± 2.76	**98.66** ± 0.73	88.88 ± 8.16	90.33 ± 3.84	94.27 ± 2.14	96.19 ± 2.10	97.11 ± 1.42
13	97.04 ± 1.20	97.13 ± 1.23	97.73 ± 1.83	97.05 ± 1.03	97.91 ± 1.08	98.45 ± 1.96	98.25 ± 1.20	97.54 ± 1.58	96.87 ± 1.88	98.89 ± 1.30	97.22 ± 1.43	99.01 ± 0.72	**99.63** ± 0.63
14	95.61 ± 3.17	95.68 ± 3.17	94.47 ± 2.65	97.39 ± 2.15	98.61 ± 0.30	96.65 ± 5.71	98.14 ± 0.68	**98.99** ± 0.43	95.49 ± 3.27	98.40 ± 0.43	96.24 ± 1.71	96.47 ± 3.10	97.55 ± 0.70
15	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	99.95 ± 0.09	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00
16	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	99.94 ± 0.18	99.77 ± 0.41	99.59 ± 0.82	**100.00** ± 0.00	99.94 ± 0.18	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00
17	99.75 ± 0.33	99.70 ± 0.35	99.95 ± 0.07	99.78 ± 0.22	99.41 ± 0.49	99.79 ± 0.28	99.97 ± 0.05	99.76 ± 0.31	99.82 ± 0.16	**99.99** ± 0.03	99.85 ± 0.17	99.91 ± 0.19	99.88 ± 0.17
18	95.73 ± 2.35	95.66 ± 2.31	93.57 ± 3.76	98.58 ± 0.84	99.08 ± 0.59	99.29 ± 0.69	98.89 ± 0.89	**99.93** ± 0.13	99.61 ± 0.46	99.68 ± 0.93	95.62 ± 1.91	98.92 ± 0.90	99.09 ± 0.91
19	96.32 ± 2.73	96.53 ± 2.67	99.26 ± 1.00	93.79 ± 3.08	97.58 ± 2.28	**100.00** ± 0.00	99.38 ± 0.76	**100.00** ± 0.00	98.41 ± 0.98	99.22 ± 1.44	96.84 ± 2.67	99.16 ± 1.09	99.55 ± 0.83
OA	95.65 ± 0.70	95.51 ± 0.80	96.45 ± 0.33	93.95 ± 1.28	97.37 ± 0.56	98.94 ± 0.78	97.49 ± 0.56	98.35 ± 1.83	96.95 ± 0.86	97.96 ± 0.60	96.06 ± 0.65	99.10 ± 0.38	**99.37** ± 0.12
AA	96.77 ± 0.46	96.68 ± 0.49	97.02 ± 0.26	95.04 ± 0.88	97.28 ± 0.19	99.08 ± 0.45	98.03 ± 0.33	98.36 ± 1.40	97.33 ± 0.59	97.95 ± 0.42	97.05 ± 0.31	99.15 ± 0.25	**99.44** ± 0.14
κ	0.950 ± 0.008	0.948 ± 0.009	0.959 ± 0.004	0.930 ± 0.015	0.970 ± 0.007	0.988 ± 0.009	0.971 ± 0.006	0.981 ± 0.021	0.965 ± 0.010	0.976 ± 0.007	0.955 ± 0.007	0.990 ± 0.004	**0.993** ± 0.001

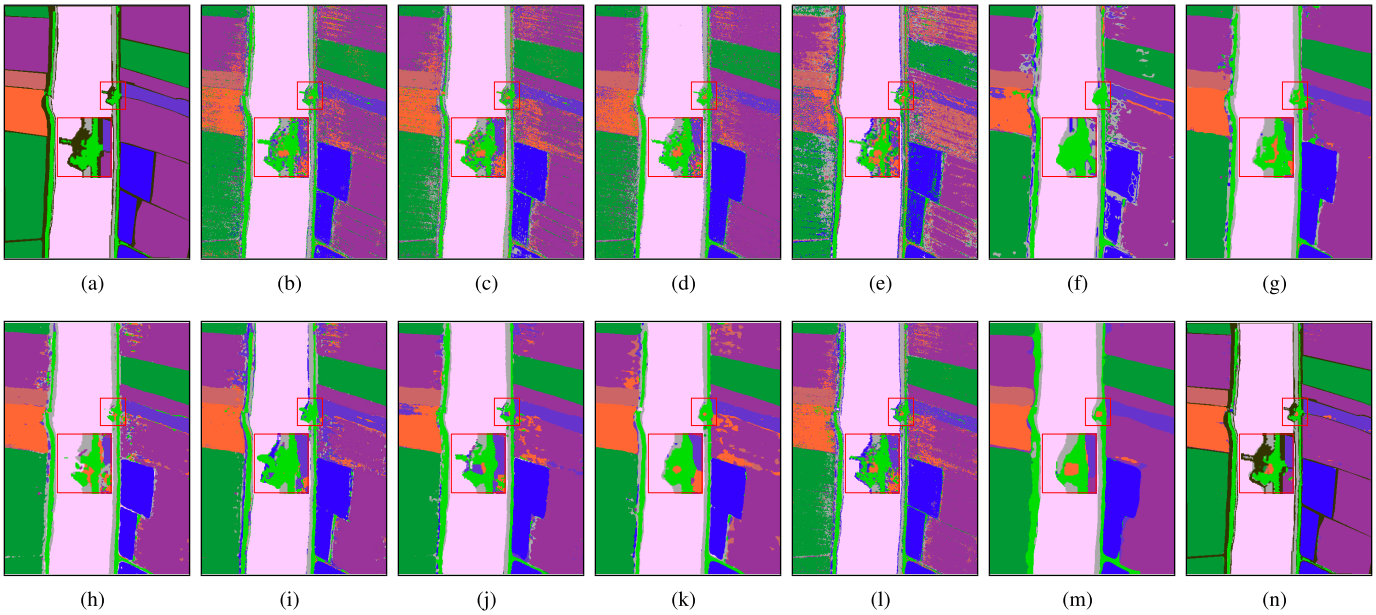


Fig. 9. Classification maps for different methods on the LK dataset. In each subfigure, a zoomed-in subregion has been added to the center to highlight performance on local structures. (a) Ground truth. (b) SVM. (c) PCA. (d) LDA. (e) ViT. (f) SpFormer. (g) GSC-ViT. (h) SSFTT. (i) DSNet. (j) MAGS. (k) SimHSIC. (l) FPCA. (m) FDDA. (n) ORF-NDA.

the distribution characteristics of land covers. The enhanced visual comparisons of zoomed-in regions clearly show that our ORF-NDA method produces more accurate and homogeneous classification areas, and fewer speckles compared to other approaches, particularly in this complex scene with highly scattered land covers.

To further demonstrate the effectiveness and advantages of the proposed ORF-NDA, we specifically test all the above methods on a more challenging HSI dataset, namely, HoustonU2013. In addition to the complexity of the HoustonU2013 scene (cloud occlusion and dispersed distribution of land covers), the selection of training samples was meticulously planned to ensure they were unevenly distributed in each region. It thus did not lead to spatial oversmoothing of the classification map. At the same time, it also makes it difficult to extract sufficiently effective spatial features with simple operations such as convolution or mean filtering. Therefore, the utilization of spectral information is crucial for this HSI dataset. Table VI presents the classification accuracies for individual classes, OA, AA, and κ for various methods on the HoustonU2013 dataset, using the training samples outlined in Table I. The proposed ORF-NDA achieved outstanding classification accuracies, with an OA of 88.87%, an AA of 90.59%, and a κ of 0.880. SSFTT and MAGS with patchwise input also achieved very competitive classification results. This illustrates the advantages of the two methods in

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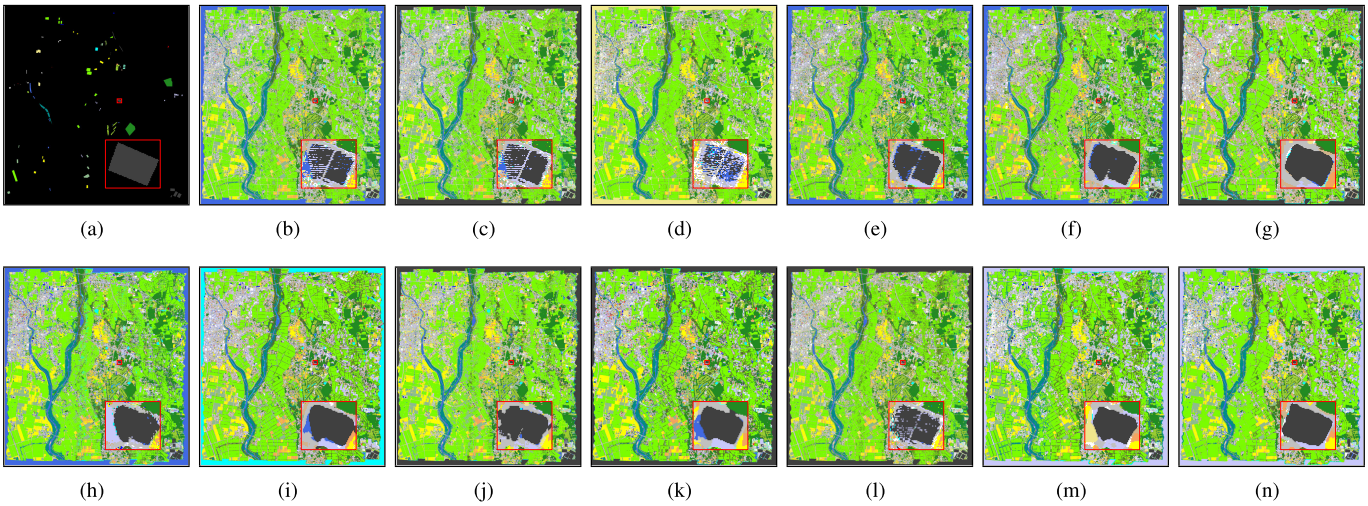


Fig. 10. Classification maps for different methods on the CH dataset. In each subfigure, a zoomed-in subregion has been added at the bottom-right to highlight performance on local structures. (a) Ground truth. (b) SVM. (c) PCA. (d) LDA. (e) ViT. (f) SpFormer. (g) GSC-ViT. (h) SSFTT. (i) DSNet. (j) MAGS. (k) SimHSIC. (l) FPCA. (m) FDDA. (n) ORF-NDA.

TABLE VI

CLASSIFICATION ACCURACIES ON THE HOUSTONU2013 DATASET OBTAINED BY DIFFERENT METHODS. THE BEST ACCURACIES ARE SHOWN IN BOLD. ALL VALUES ARE IN PERCENTAGE EXCEPT FOR κ

Class	Conventional Algorithms			Deep Learning Algorithms							Functional Data Learning Algorithms		
	SVM	**PCA**	**LDA**	**ViT**	**SpFormer**	**GSC-ViT**	**SSFTT**	**DSNet**	**MAGS**	**SimHSIC**	**FPCA**	**FDDA**	**ORF-NDA**
1	83.10	83.00	81.58	82.05	84.81	81.95	84.81	84.43	85.69	**86.30**	83.00	79.49	81.93
2	82.99	82.71	86.18	78.20	99.44	82.36	92.42	**99.53**	95.37	97.99	82.71	84.77	98.56
3	99.80	99.80	**100.00**	**100.00**	98.61	96.87	98.28	95.25	99.71	**100.00**	99.80	99.21	**100.00**
4	91.29	90.15	98.11	92.71	96.78	91.39	96.06	**99.53**	92.52	99.11	98.48	75.85	93.25
5	98.30	97.54	96.88	96.69	**100.00**	99.89	**100.00**	**100.00**	99.84	**100.00**	97.92	**100.00**	99.60
6	99.30	94.41	99.30	95.10	97.90	92.94	**100.00**	92.31	98.15	91.75	99.30	95.10	**100.00**
7	80.60	75.93	85.26	74.25	**89.37**	84.00	84.38	86.75	84.23	87.20	85.45	68.94	78.00
8	43.40	41.69	54.51	38.18	**83.95**	68.90	70.10	76.92	69.53	67.42	47.39	79.30	78.86
9	78.09	74.88	67.42	68.18	75.73	81.11	**91.45**	73.75	88.18	81.16	72.24	82.06	88.10
10	54.73	53.96	46.72	50.19	53.28	46.24	68.13	58.88	60.47	69.29	58.49	90.35	**91.20**
11	73.34	73.91	62.05	62.52	83.78	74.07	84.70	88.52	**88.99**	88.73	74.95	64.33	70.93
12	81.36	75.02	70.99	53.03	79.63	81.82	92.38	81.27	**94.65**	69.69	73.97	94.04	89.54
13	77.89	78.95	61.40	57.89	75.44	88.67	**91.47**	87.02	85.07	83.44	67.72	83.86	88.91
14	98.79	97.57	99.60	99.60	**100.00**	98.70	**100.00**	95.55	**100.00**	**100.00**	99.19	**100.00**	**100.00**
15	97.46	96.62	98.31	98.31	**100.00**	99.94	99.24	97.67	99.39	99.38	97.46	**100.00**	**100.00**
OA	79.27	77.58	77.45	72.67	86.17	81.55	88.41	86.28	87.84	86.61	79.68	83.86	**88.87**
AA	82.70	81.08	80.55	76.46	87.91	84.59	90.23	87.82	89.45	88.10	82.54	86.49	**90.59**
κ	0.776	0.759	0.756	0.706	0.850	0.801	0.875	0.851	0.868	0.855	0.780	0.825	**0.880**

spectral feature extraction and can effectively mine long-term sequence features of high-dimensional spectral information by the Transformer and LSTM. In particular, the proposed ORF-NDA has more advantages for high-dimensional data analysis due to its stronger feature expression and nonlinear feature extraction abilities. SimHSIC, DSNet, and SpFormer also achieve very good classification accuracy because they enhance the utilization of spectral information. Except for the above advanced deep learning algorithm, the FDDA algorithm achieved better results than the remaining comparison methods, which once again illustrates the advantages of functional data learning methods for high-dimensional data processing and analysis. Comparing GSC-ViT and SpFormer, although both use patches as input, SpFormer is more effective, indicating its advantages in feature extraction of long-term spectral data. The functional benchmark algorithm, FPCA, outperforms the three traditional machine learning algorithms (SVM, PCA, and LDA) as well as the benchmark deep learning algorithms (ViT). This highlights the superiority of functional data learning algorithms in processing hyperspectral data, particularly in their ability to extract effective features of spectral information.

Fig. 11 showcases the classification maps and zoomed-in regions of the HoustonU2013 scene, produced by the various methods outlined in Table VI. Visually, the proposed ORF-NDA method yields the most accurate classification map, depicted in Fig. 11(n), particularly in the right section of the image where no training samples exist due to cloud coverage. Despite the lack of training data in this area, the proposed ORF-NDA method successfully delineates some land covers with clear boundaries. Although the zoomed-in region on the right side of the map, Fig. 8(n), has a small number of misclassified pixels, the edge in the center is preserved better than other methods, and there is the fewest misclassification for the target class (Class 10 in Table VI). In contrast, the

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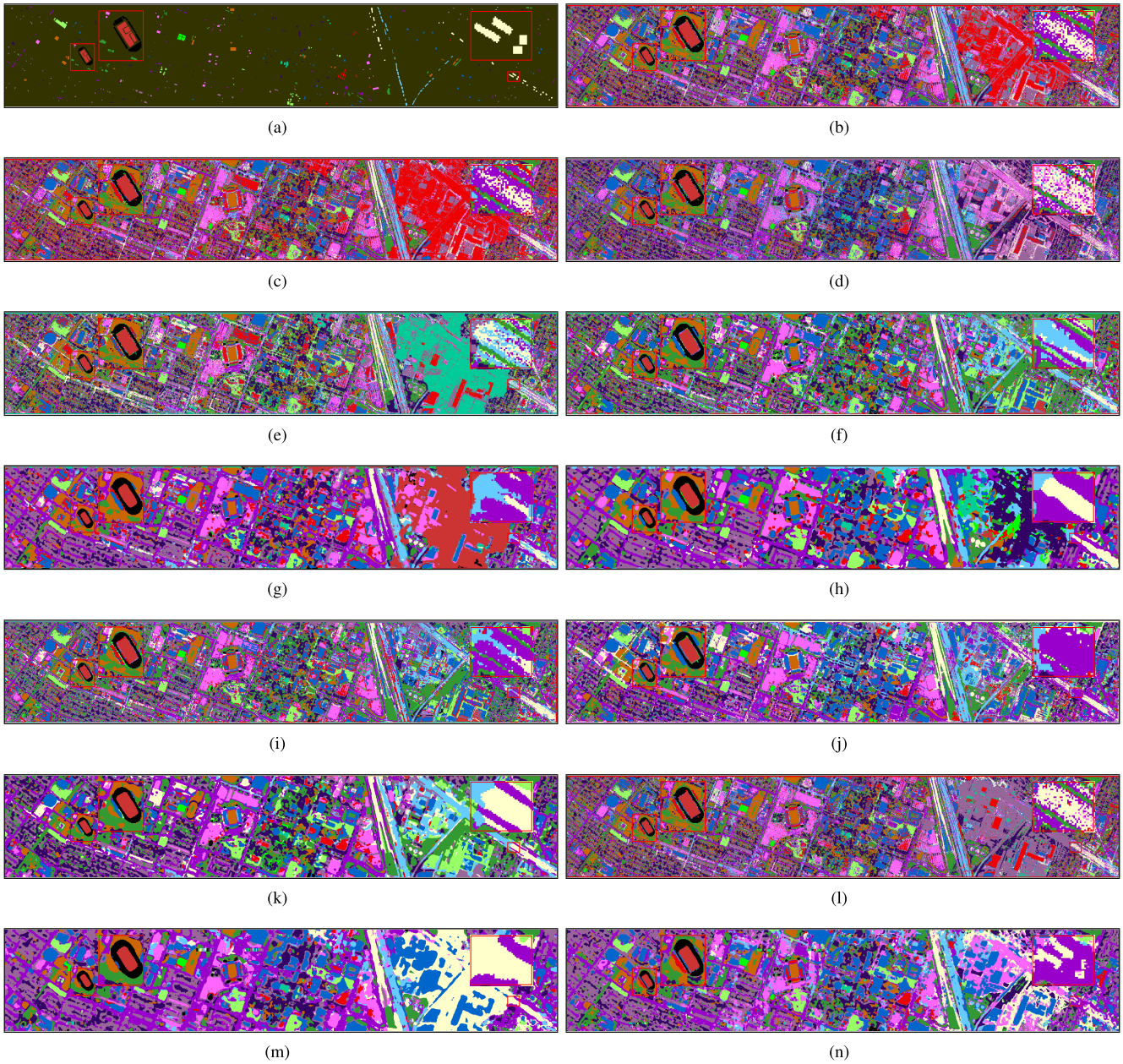


Fig. 11. Classification maps for different methods on the HoustonU2013 dataset. In each subfigure, two zoomed-in subregions have been added on the left and right to highlight performance on local structures. (a) Ground truth. (b) SVM. (c) PCA. (d) LDA. (e) ViT. (f) SpFormer. (g) GSC-ViT. (h) SSFTT. (i) DSNet. (j) MAGS. (k) SimHSIC. (l) FPCA. (m) FDDA. (n) ORF-NDA.

classification maps from other methods, such as Fig. 11(b)–(e), (g) and (h), and (k)–(m), are heavily impacted by cloud interference, resulting in visible shadow artifacts. This causes most of the pixels in this area to be uniformly labeled as a certain class, resulting in significant spatial oversmoothing. Although MAGS, DSNet, and SpFormer attempted to classify the land covers in the shadow area by strengthening the use of spectral information, they still produced incorrect classifications, resulting in decreased accuracy.

C. Computational Complexity Analysis

We have conducted a comparative evaluation of the proposed ORF-NDA against competing algorithms in terms of runtime performance in this section. However, it is essential to note that the proposed ORF-NDA was implemented on a CPU using MATLAB with parpool for parallel computing. This greatly improves the speed of training. In contrast, the other algorithms were executed on an NVIDIA RTX 4090D (24GB GPU) with PyTorch. This hardware setup inherently favors GPU-accelerated deep learning methods.

The computational time of the proposed ORF-NDA is related to the number of samples (n), the number of classes (C), the number of basis functions (K), and the number of spectral bands (d), where the computational efficiency of solving functional data fitting model is directly related to the number of samples, the number of basis functions, and the number of spectral bands, while the computational efficiency

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TABLE VII
COMPUTATIONAL TIME COST ON THE FOUR HSI DATASETS OBTAINED BY DIFFERENT METHODS

Datasets	Time(s)	Conventional Algorithms			Deep Learning Algorithms							Functional Data Learning Algorithms		
		SVM	**PCA**	**LDA**	**ViT**	**SpFormer**	**GSC-ViT**	**SSFTT**	**DSNet**	**MAGS**	**SimHSIC**	**FPCA**	**FDDA**	**ORF-NDA**
IP	Training	38.70	10.90	**4.38**	372.09	117.21	116.74	13.12	516.37	47.31	227.93	52.61	4.66	16.87
	Test	1.94	0.61	**0.21**	1.89	10.25	2.28	1.61	3.59	7.80	8.58	0.44	0.21	0.32
PU	Training	7.57	5.90	**0.95**	977.27	935.94	324.81	5.59	264.26	33.07	561.68	94.16	1.19	12.61
	Test	5.17	2.54	0.58	16.68	110.19	29.07	6.68	9.89	64.83	61.55	1.96	0.56	**0.38**
LK	Training	19.23	19.65	**1.20**	922.05	1197.81	1331.66	6.97	263.92	32.68	1998.79	317.04	1.36	27.30
	Test	13.21	3.02	1.47	23.81	42.02	38.86	7.33	16.98	59.09	71.16	1.85	0.61	**0.56**
CH	Training	35.52	14.71	**6.57**	1516.08	1441.81	799.98	13.83	246.91	80.39	440.44	8.96	6.58	173.96
	Test	338.88	121.71	68.93	145.48	110.86	713.77	1353.17	1841.25	2270.63	1840.51	90.16	52.63	**27.69**
HoustonU2013	Training	235.88	88.12	30.81	890.78	392.25	432.47	**8.49**	181.95	168.59	594.05	288.53	39.06	75.55
	Test	83.94	32.94	13.74	57.56	6.65	83.75	39.34	46.87	181.14	159.13	18.54	5.67	**3.48**

TABLE VIII
ABLATION STUDIES OF THE ORF-NDA METHOD AND THE MEAN FILTERING OPERATION ON THE FIVE HSI DATASETS. THE EVALUATION METRIC IS OA (%)

Mean filtering	**ORF-NDA**	**IP**	**PU**	**LK**	**CH**	**HoustonU2013**
×	×	71.97	74.66	91.78	95.65	79.27
✓	×	88.62	90.10	95.97	97.91	81.93
×	✓	75.09	75.78	94.59	96.99	82.93
✓	✓	**94.03**	**92.56**	**98.46**	**99.37**	**88.87**

of solving NDA model is related to the number of basis functions and the number of classes. Despite this discrepancy of hardware and software setup, we have included the absolute runtime measurements in the experimental results (Table VII) to provide a reference for computational cost.

While the proposed ORF-NDA incurs a higher computational cost than PCA and LDA, it remains more efficient than most deep learning approaches across the five HSI datasets. Applying parallel computing via MATLAB's parpool, ORF-NDA rapidly solves the functional data fitting model using SAFD and computes feature projections, substantially reducing training time. Notably, while both its training and testing times increase moderately with dataset scale, ORF-NDA achieves the fastest testing speed on four of the five datasets. Overall, it outperforms most deep learning algorithms in terms of time efficiency, with only SSFTT showing good runtime performance on small datasets. However, its test time increases significantly on the large-scale CH dataset. Additionally, we emphasize that the proposed ORF-NDA is designed to reduce reliance on high-end computing resources, making it more accessible for environments that do not have GPU acceleration.

D. Ablation Experiments

Table VIII presents an ablation study evaluating the individual contributions of the ORF-NDA method and local mean filtering on five HSI datasets. When ORF-NDA is disabled, applying mean filtering significantly improves OA across five datasets, especially on the IP and PU datasets compared to no filtering. This indicates that spatial smoothing enhances classification by reducing noise and strengthening spatial coherence. When mean filtering is disabled, but ORF-NDA is applied, significant gains are observed across five datasets. This demonstrates that ORF-NDA effectively captures discriminative spectral features even without preprocessing. The highest OA values are achieved when both mean filtering and ORF-NDA are enabled. On the IP dataset, OA jumps from 71.97% to 94.03%, and on the CH dataset, it reaches 99.37%. This is a clear indication of strong synergy between the two components. Mean filtering preprocesses the data to enhance spatial consistency, while ORF-NDA leverages this refined input to extract more robust representations. The CH dataset shows the most dramatic improvement with both techniques, likely due to its high spatial resolution. The HoustonU2013 dataset, being urban and spectrally complex, benefits significantly from the two components (from 79.27% to 88.87%), highlighting the importance of noise suppression in heterogeneous scenes. The LK dataset already achieves high accuracy with ORF-NDA alone (96.99%), suggesting strong intrinsic spectral separability.

In summary, this ablation study confirms that both local mean filtering and ORF-NDA contribute significantly and complementarily to classification performance. While ORF-NDA provides powerful feature learning, mean filtering acts as an effective spatial regularizer, especially beneficial in noisy or complex environments. Their combined use leads to substantial performance gains across diverse HSI scenarios.

IV. CONCLUSION

Different from the mainstream processing mode, ORF-NDA is built in the function space with functions as the basic unit instead of data vectors. It is very suitable for processing high-dimensional, continuously changing data. An HSI can be regarded as a discrete sampling of an underlying infinite-dimensional tensor composed of spectral curves arranged in a specific space (geometric distribution shape of land covers). This means that this proposed ORF-NDA method is not only suitable for current hyperspectral image (HSI) processing but also has good generalization performance for higher-dimensional ultraspectral image processing. The proposed ORF-NDA method excels in extracting spectral features and achieves impressive classification performance

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with just simple mean filtering. Compared with the current advanced algorithms, the proposed method demonstrated advantages, especially for higher-dimensional and large-area complex datasets. In contrast, other state-of-the-art methods showed extremely unstable performance and require powerful GPU computing, performing well on some datasets and declining on other datasets, and even take significantly longer on large datasets. Despite its significant advantages, the proposed ORF-NDA has two key improvements in future work. First, GPU-based implementation and algorithmic improvements will be explored to accelerate the solution of the functional optimization model in (18), outperforming CPU-based parallel computing. Second, ORF-NDA focuses primarily on spectral feature extraction and underutilizes spatial information. Future research will address this by developing functional-paradigm spatial feature extraction algorithms and integrating them into a spectral–spatial classification framework to further enhance performance.

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